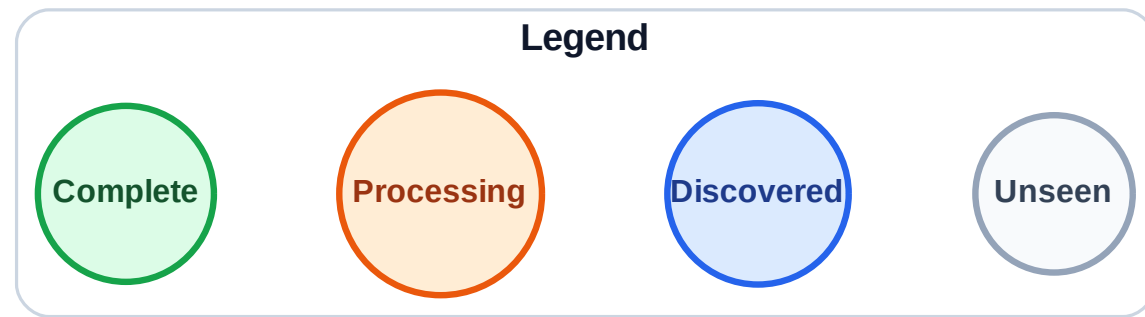


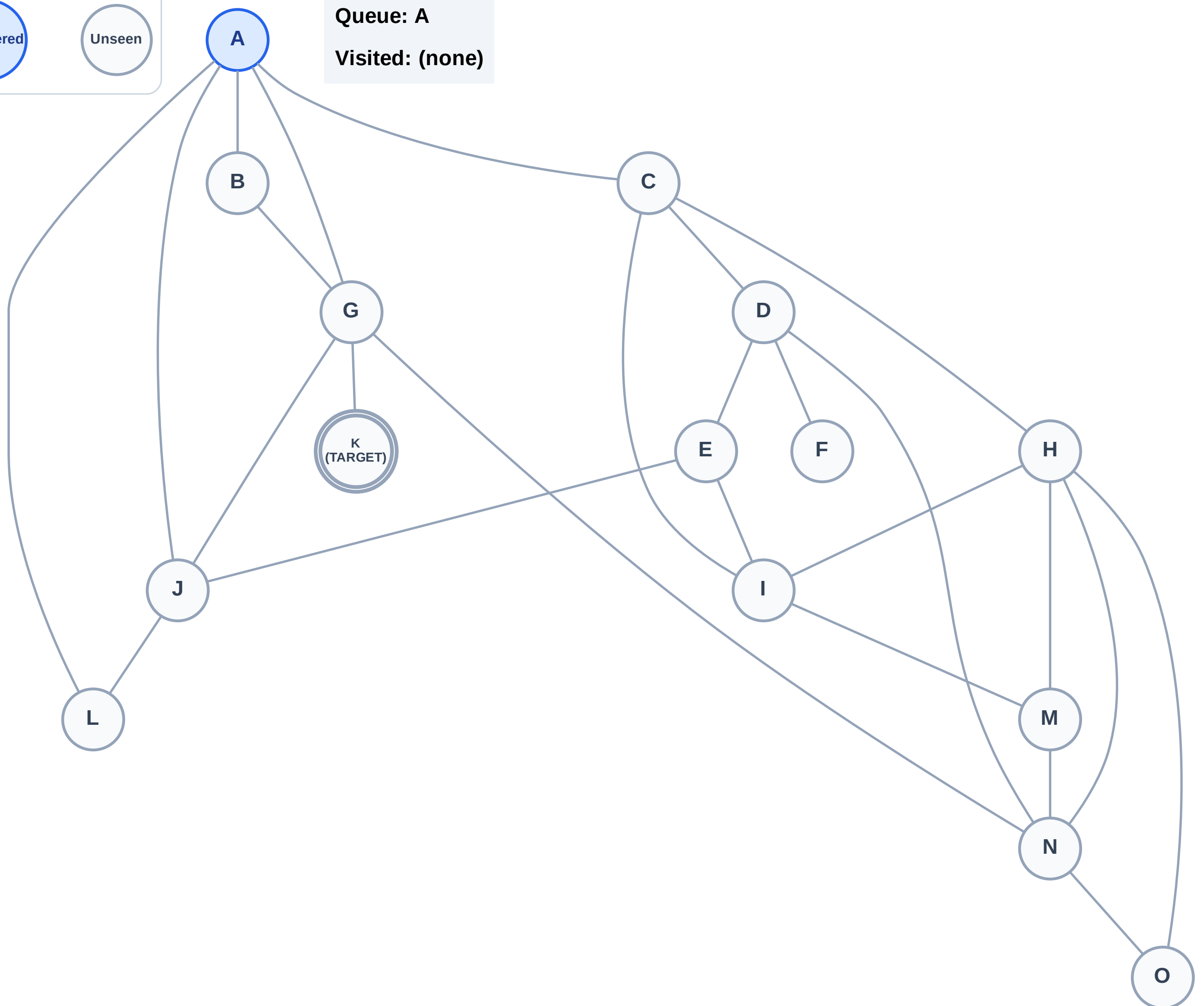
BFS Traversal

Step 1: Start at A (Seeking target node K)



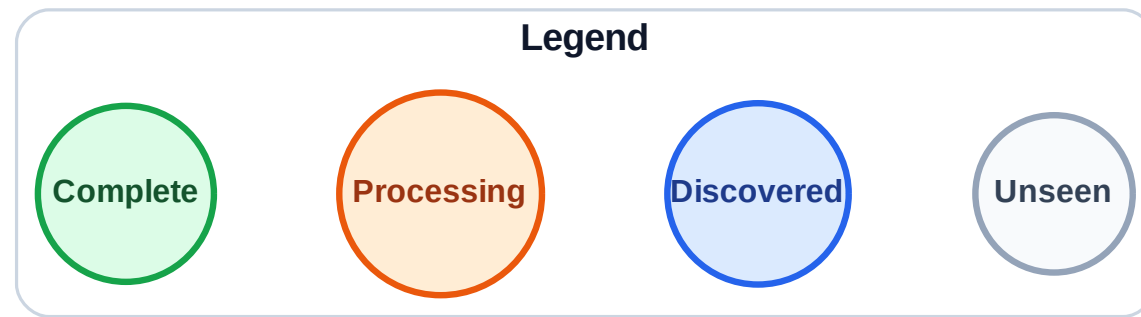
Queue: A

Visited: (none)



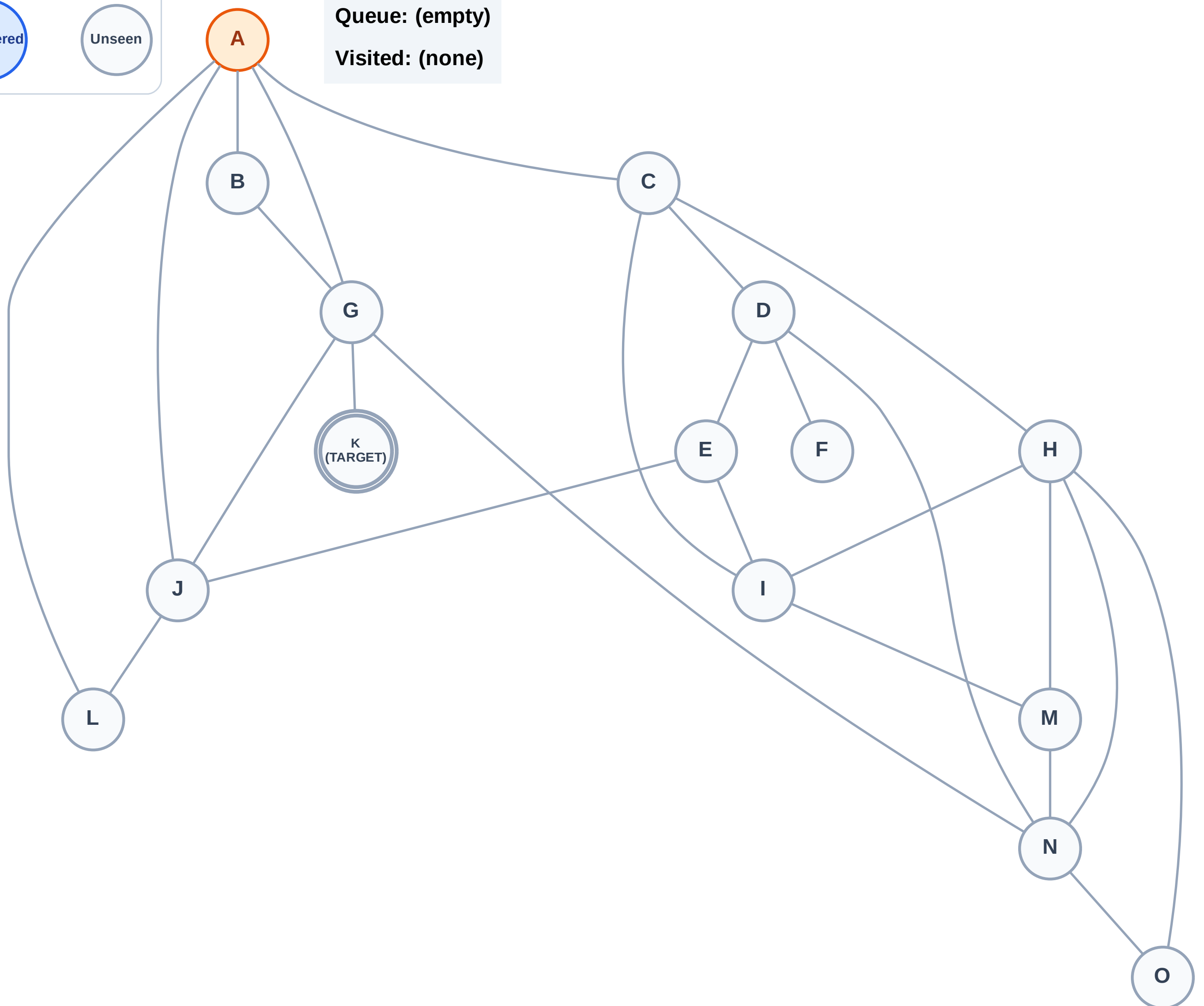
BFS Traversal

Step 2: Process node A



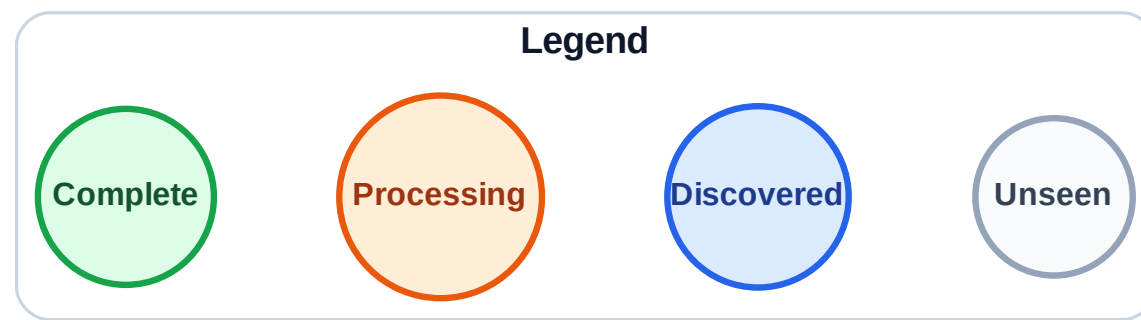
Queue: (empty)

Visited: (none)



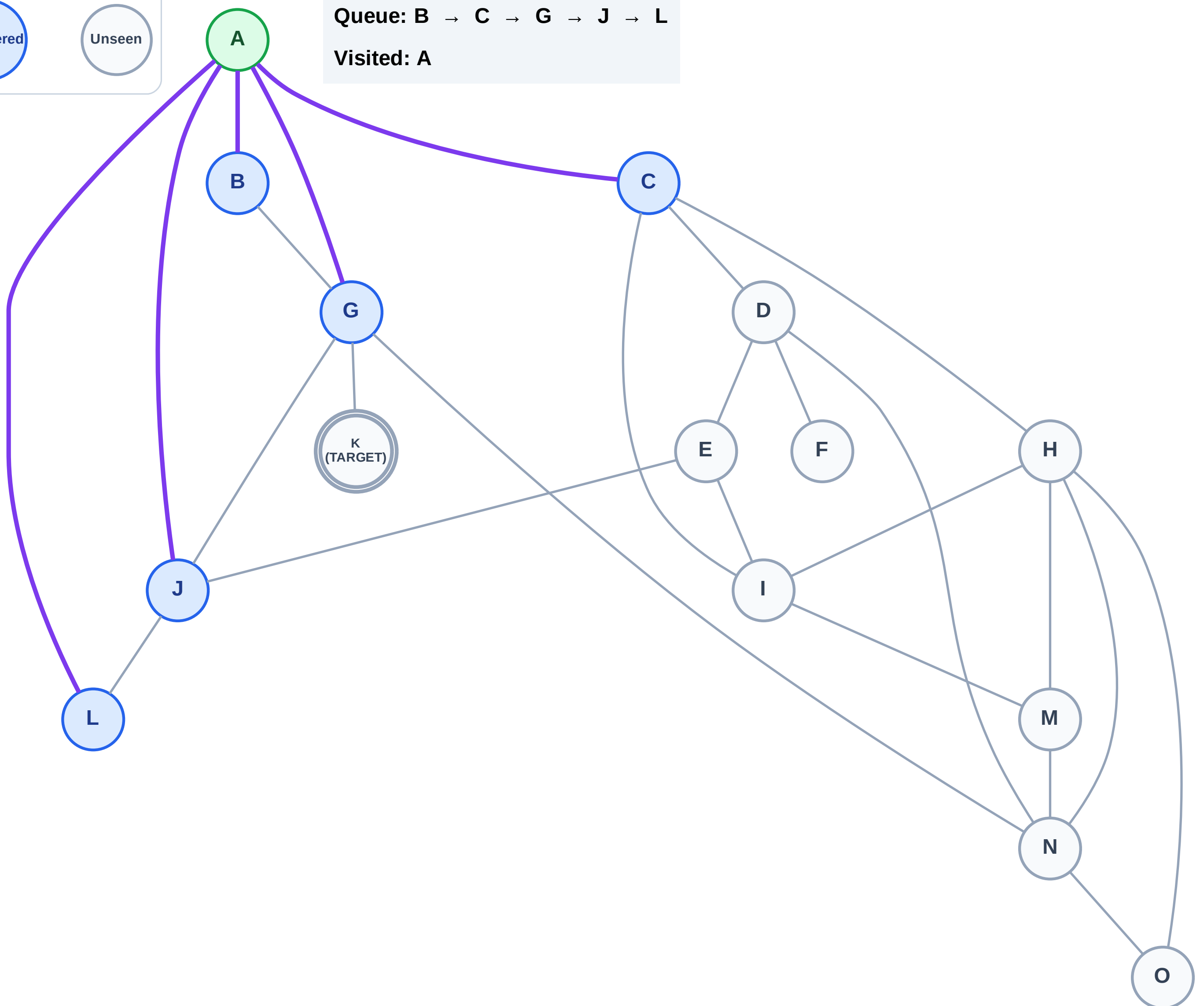
BFS Traversal

Step 3: Discover B, C, G, J, L from A



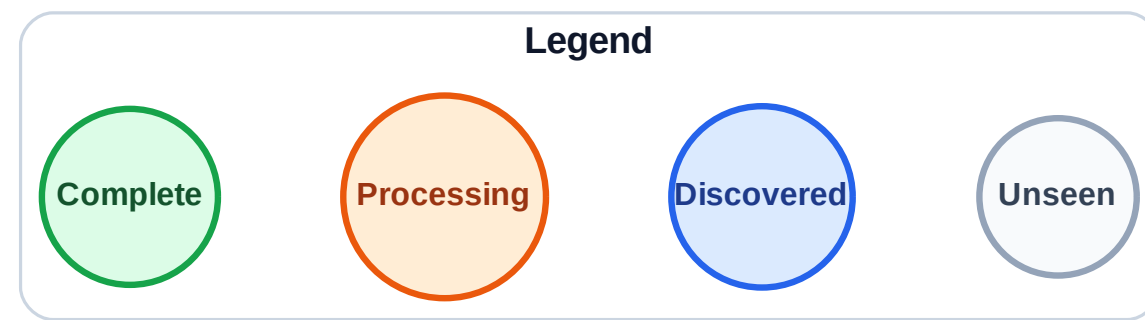
Queue: B → C → G → J → L

Visited: A



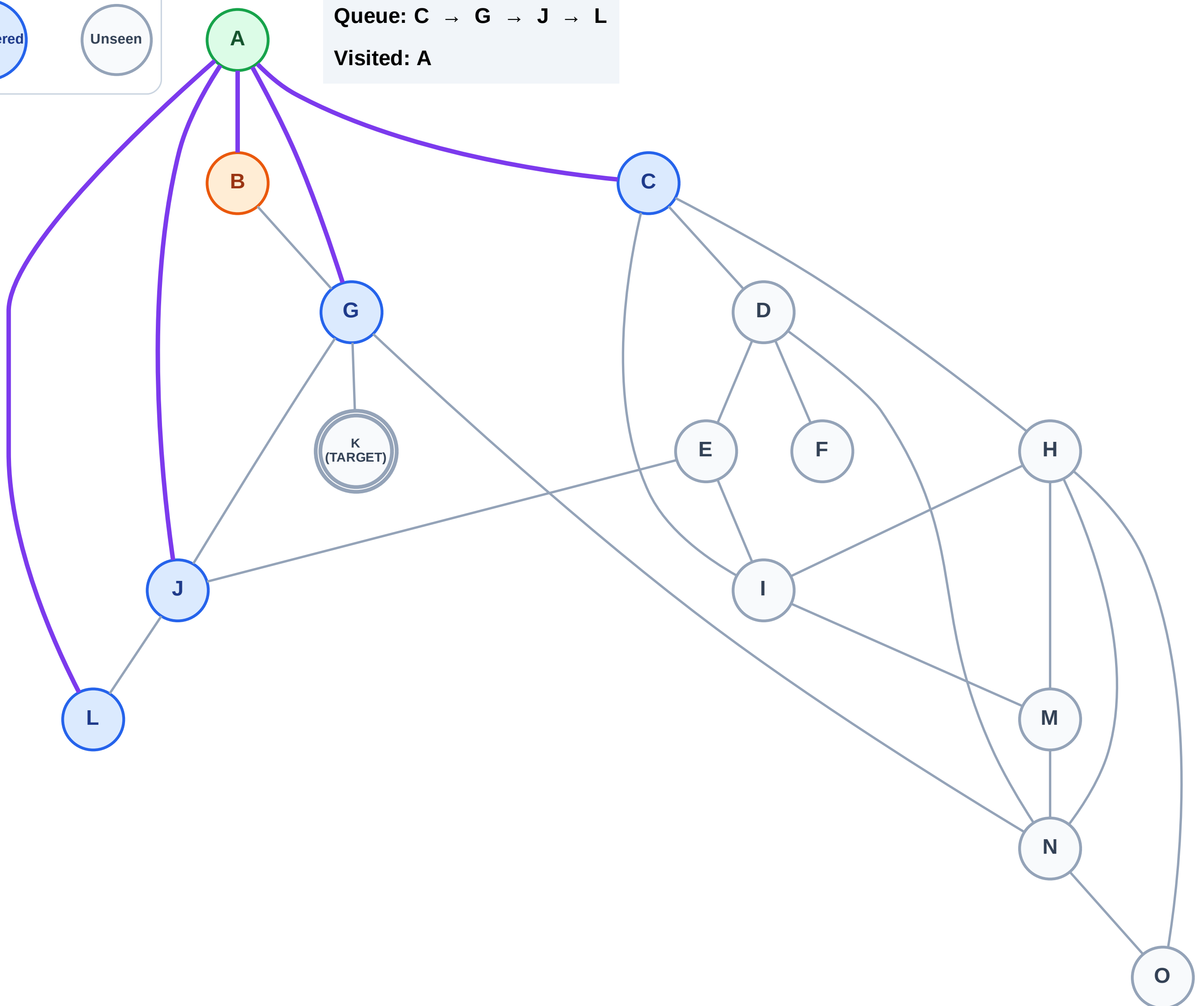
BFS Traversal

Step 4: Process node B



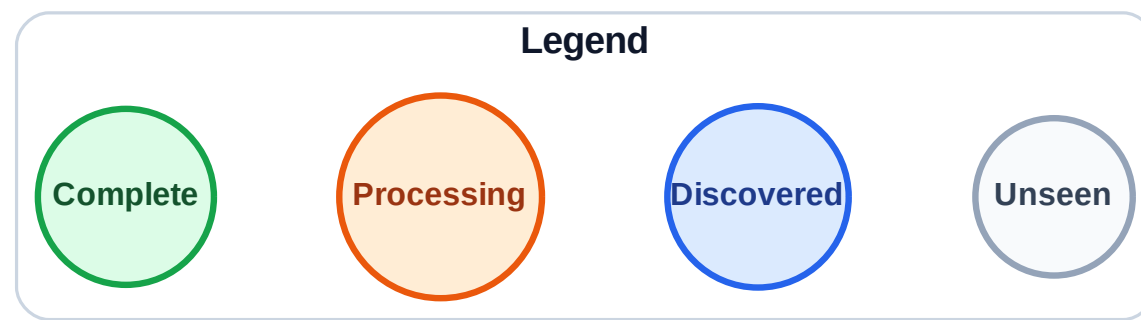
Queue: C → G → J → L

Visited: A



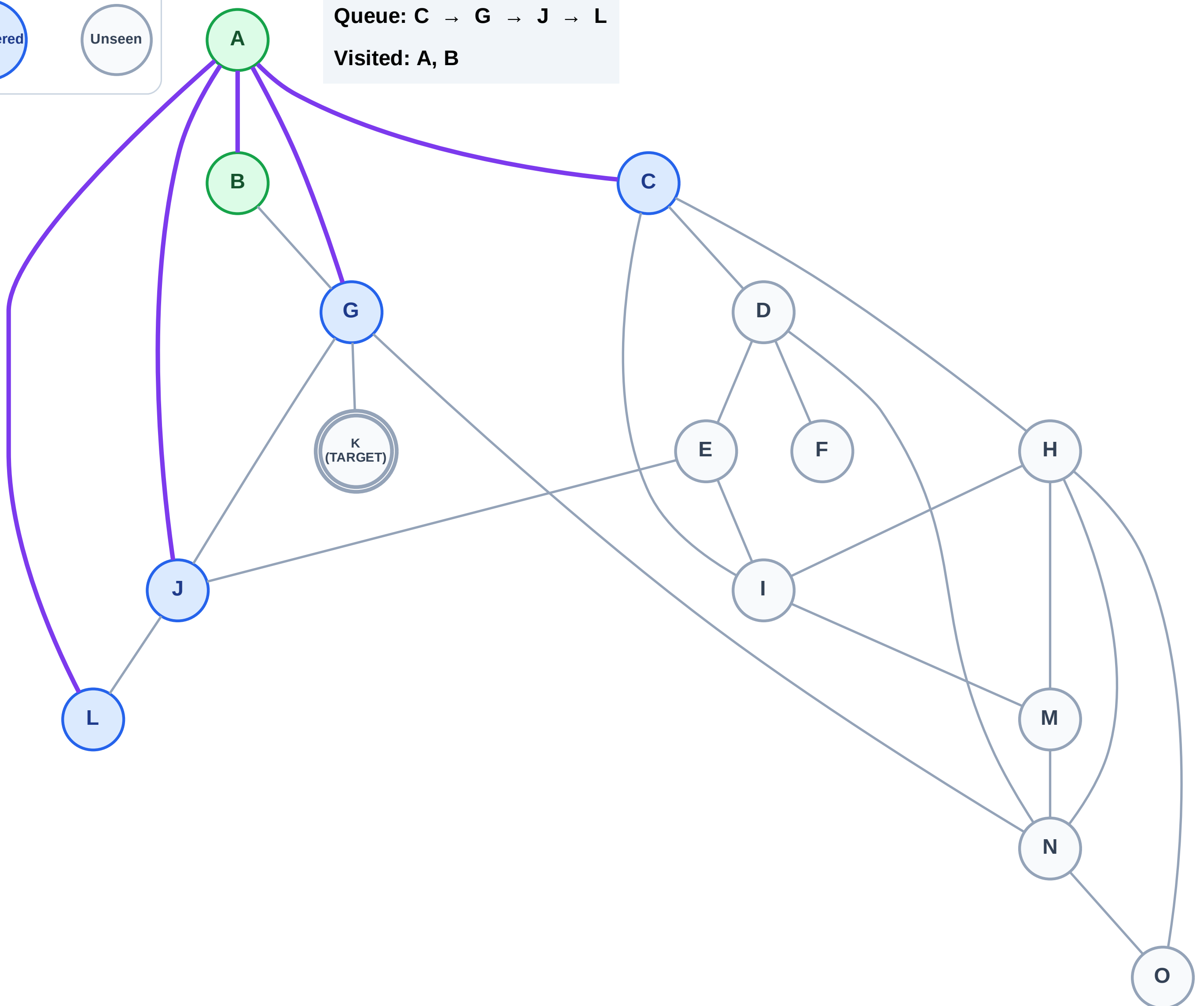
BFS Traversal

Step 5: No new neighbors from B



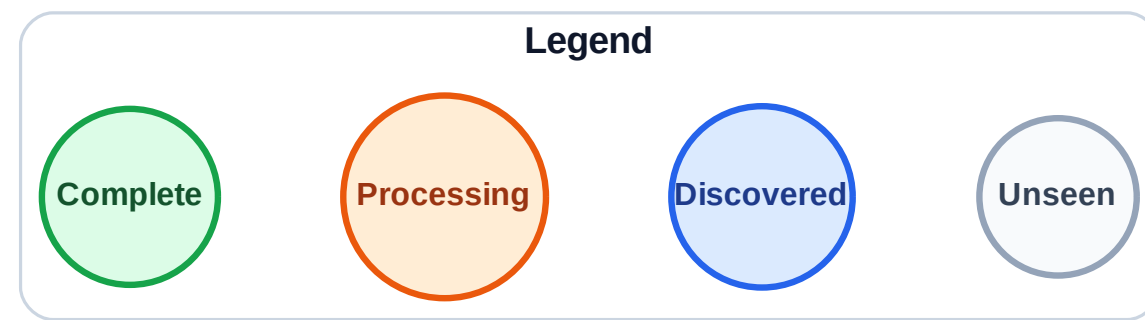
Queue: C → G → J → L

Visited: A, B



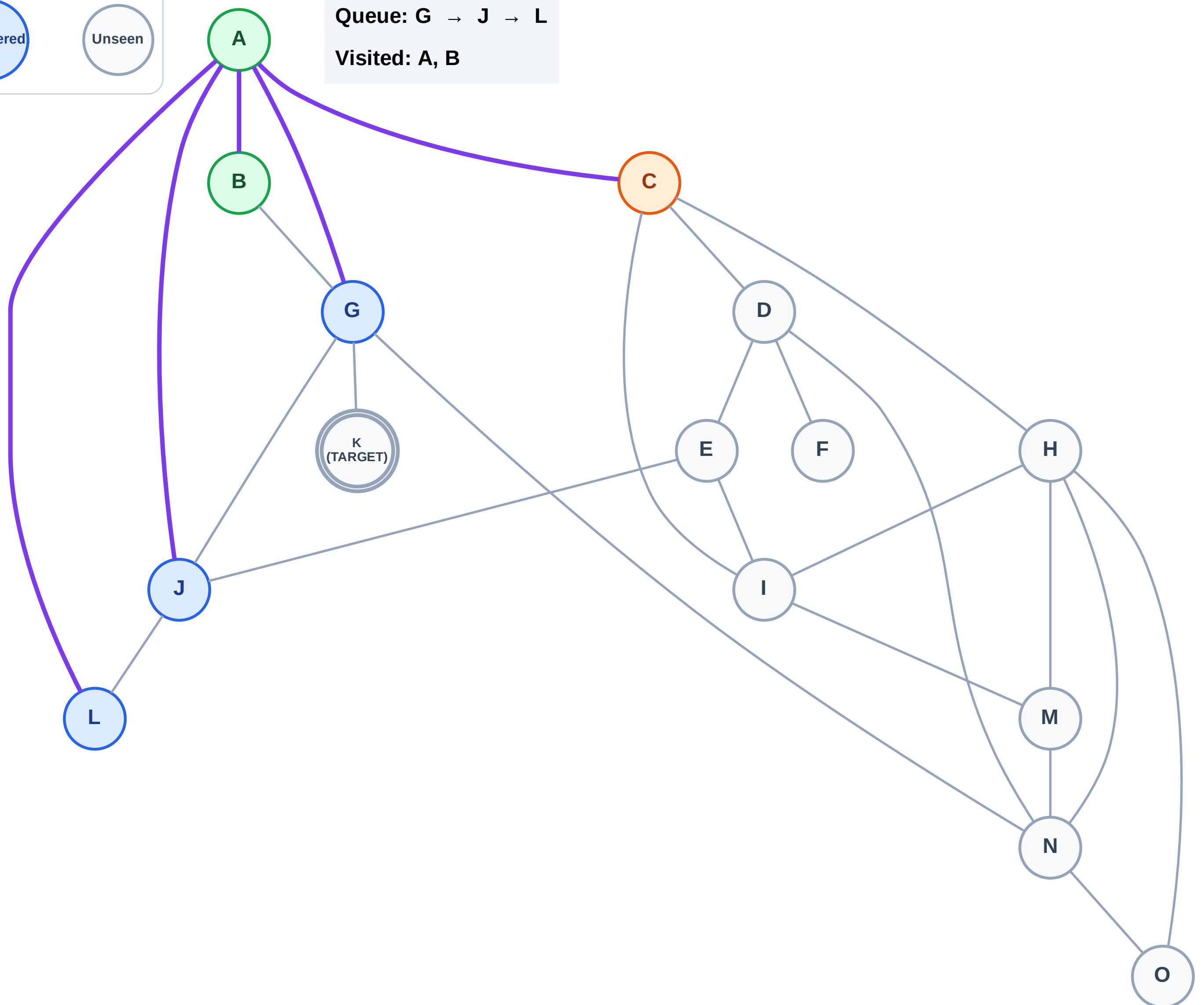
BFS Traversal

Step 6: Process node C



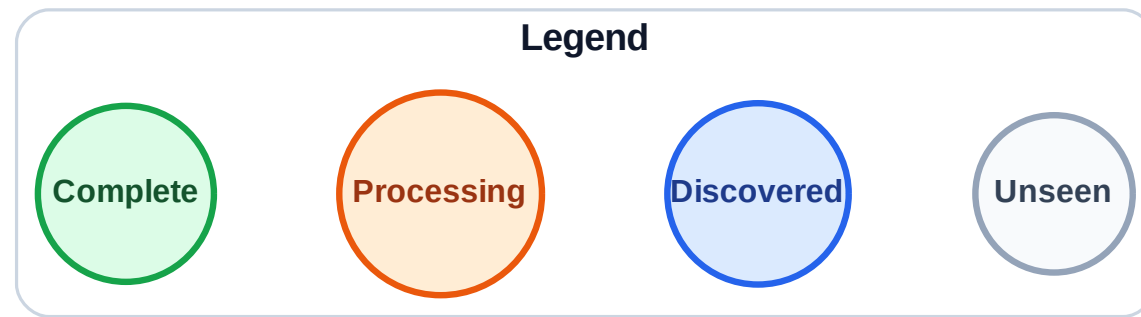
Queue: G → J → L

Visited: A, B



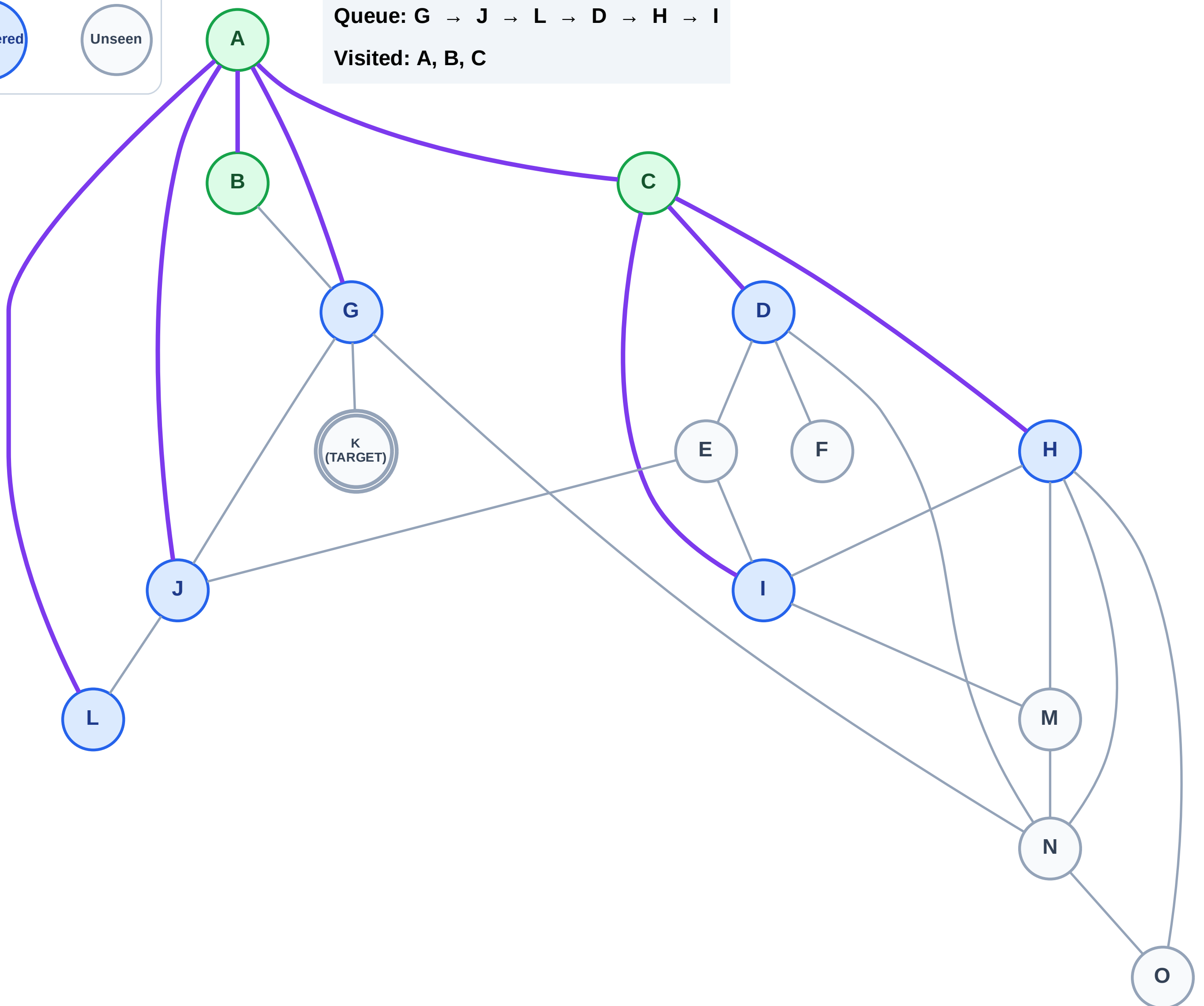
BFS Traversal

Step 7: Discover D, H, I from C



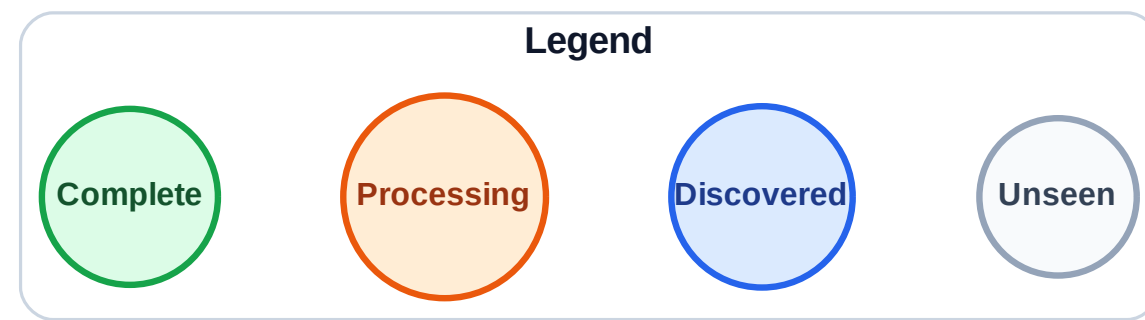
Queue: G → J → L → D → H → I

Visited: A, B, C



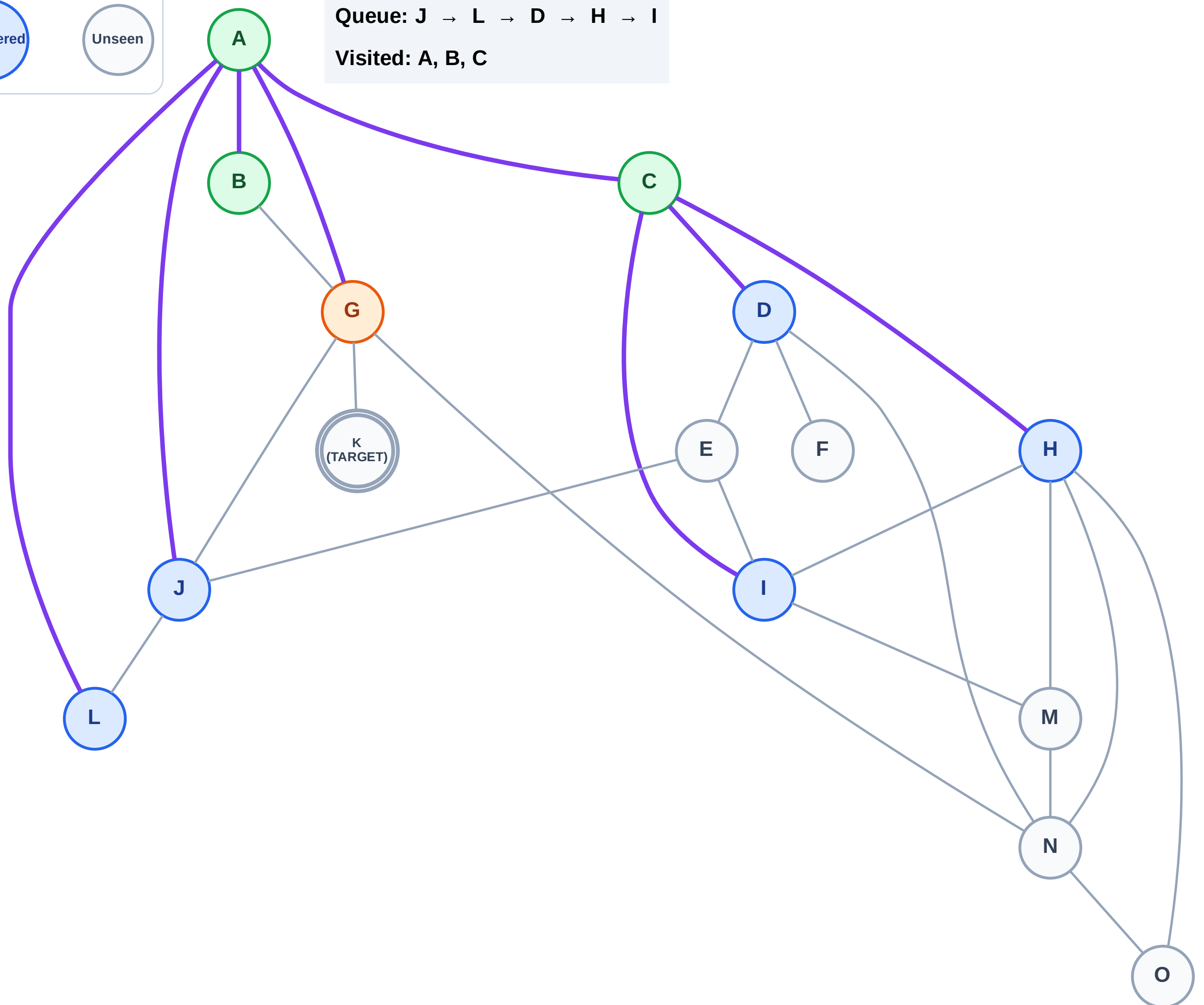
BFS Traversal

Step 8: Process node G



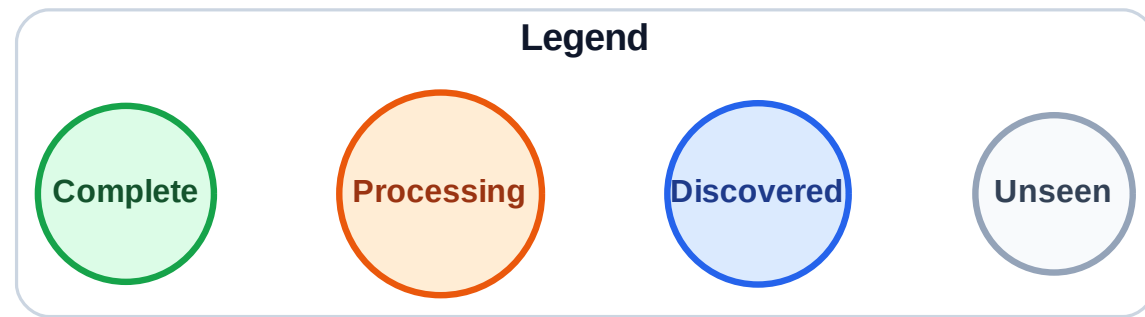
Queue: J → L → D → H → I

Visited: A, B, C



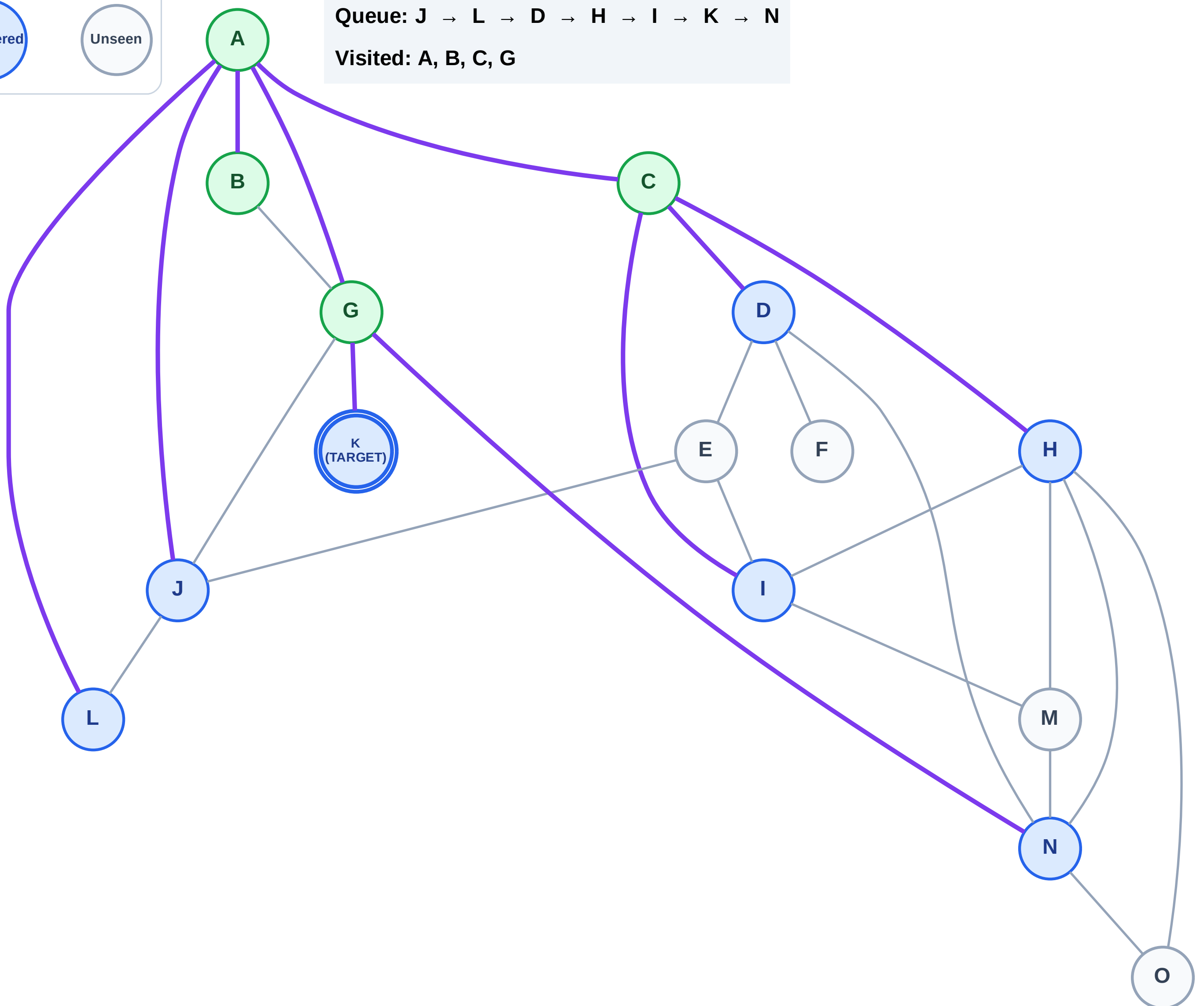
BFS Traversal

Step 9: Discover K, N from G



Queue: J → L → D → H → I → K → N

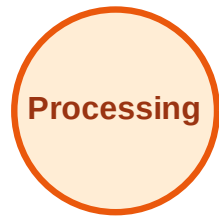
Visited: A, B, C, G



BFS Traversal

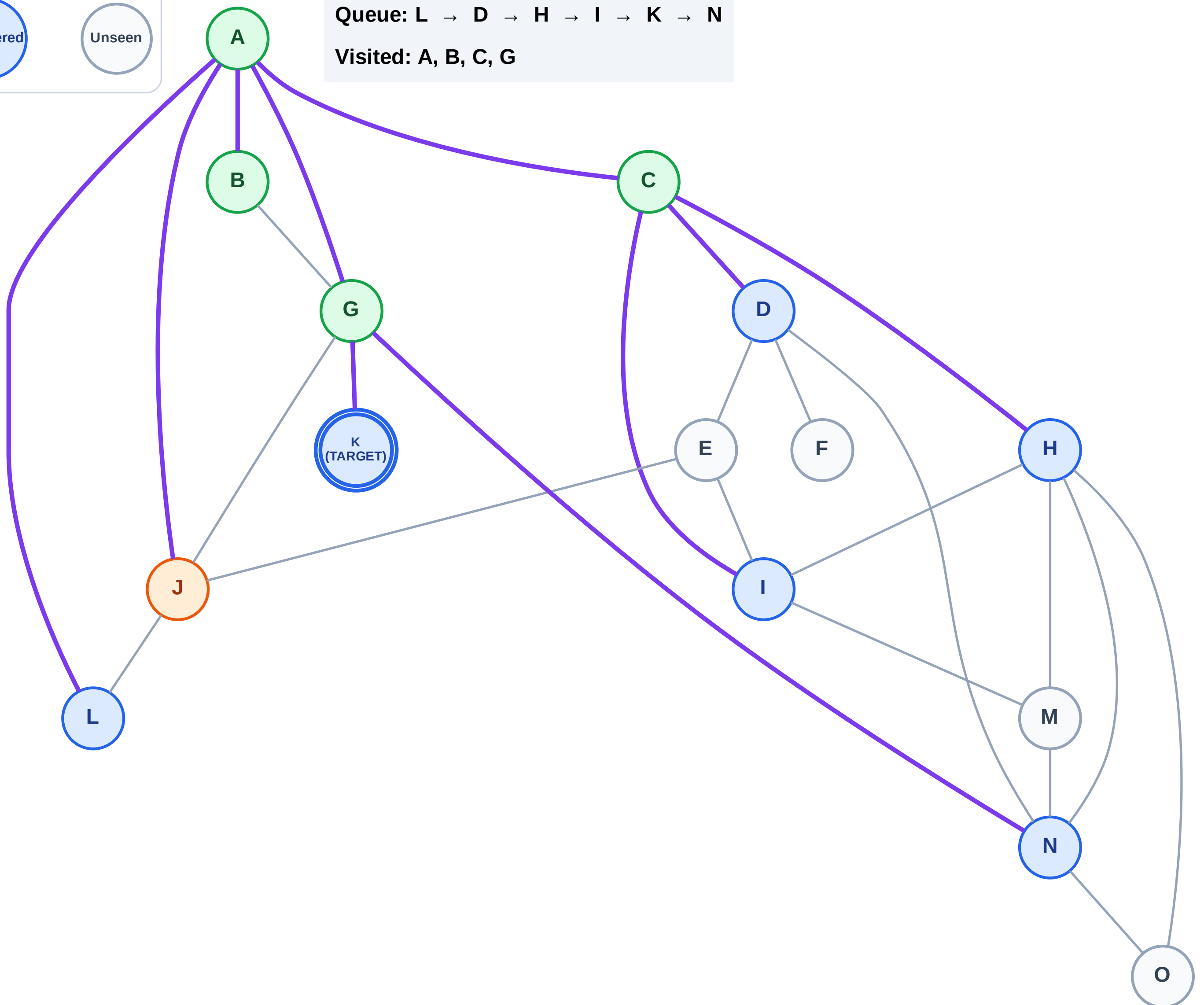
Step 10: Process node J

Legend



Queue: L → D → H → I → K → N

Visited: A, B, C, G



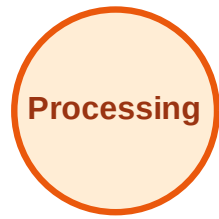
BFS Traversal

Step 11: Discover E from J

Legend



Complete



Processing



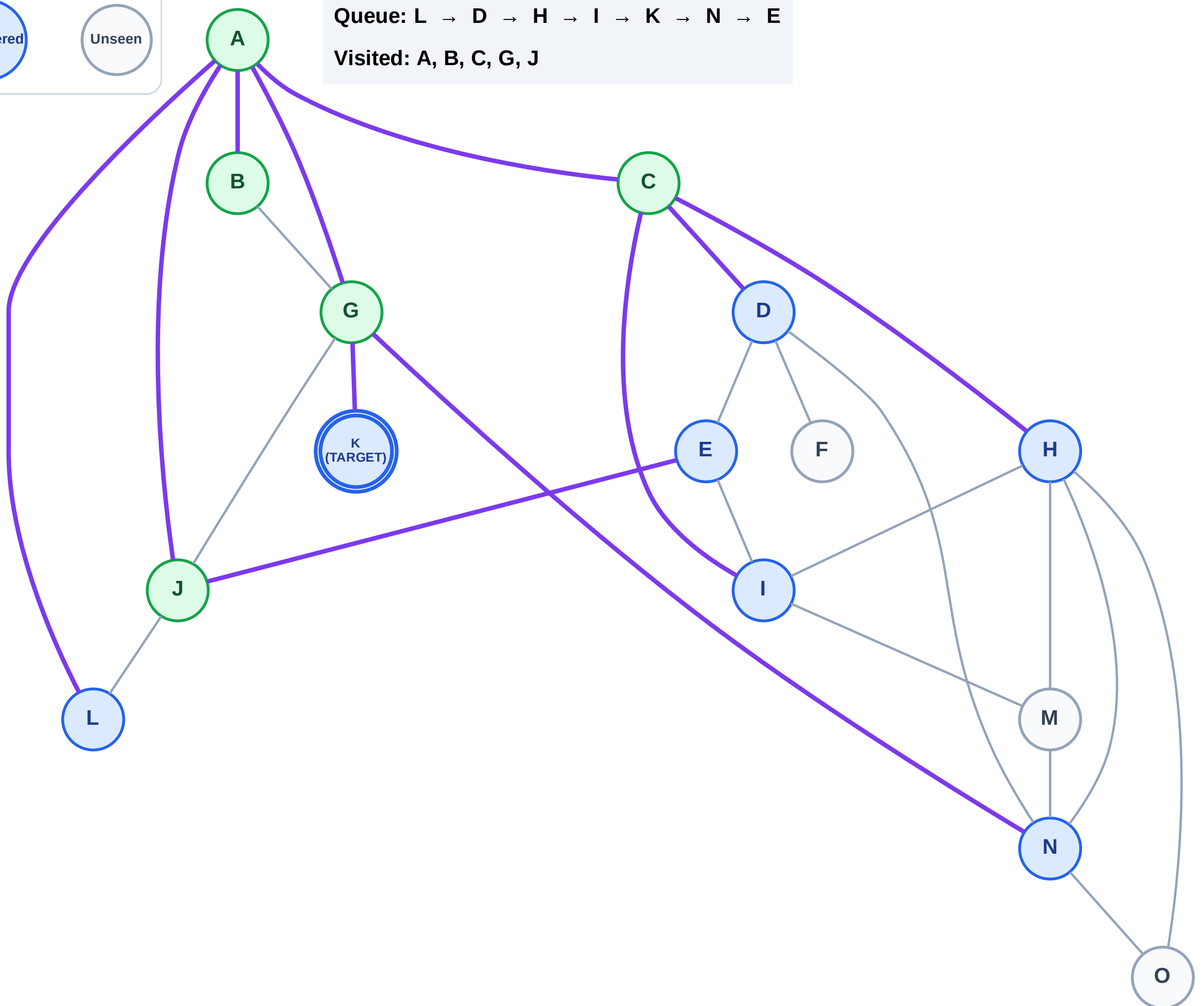
Discovered



Unseen

Queue: L → D → H → I → K → N → E

Visited: A, B, C, G, J



BFS Traversal

Step 12: Process node L

Legend

Complete

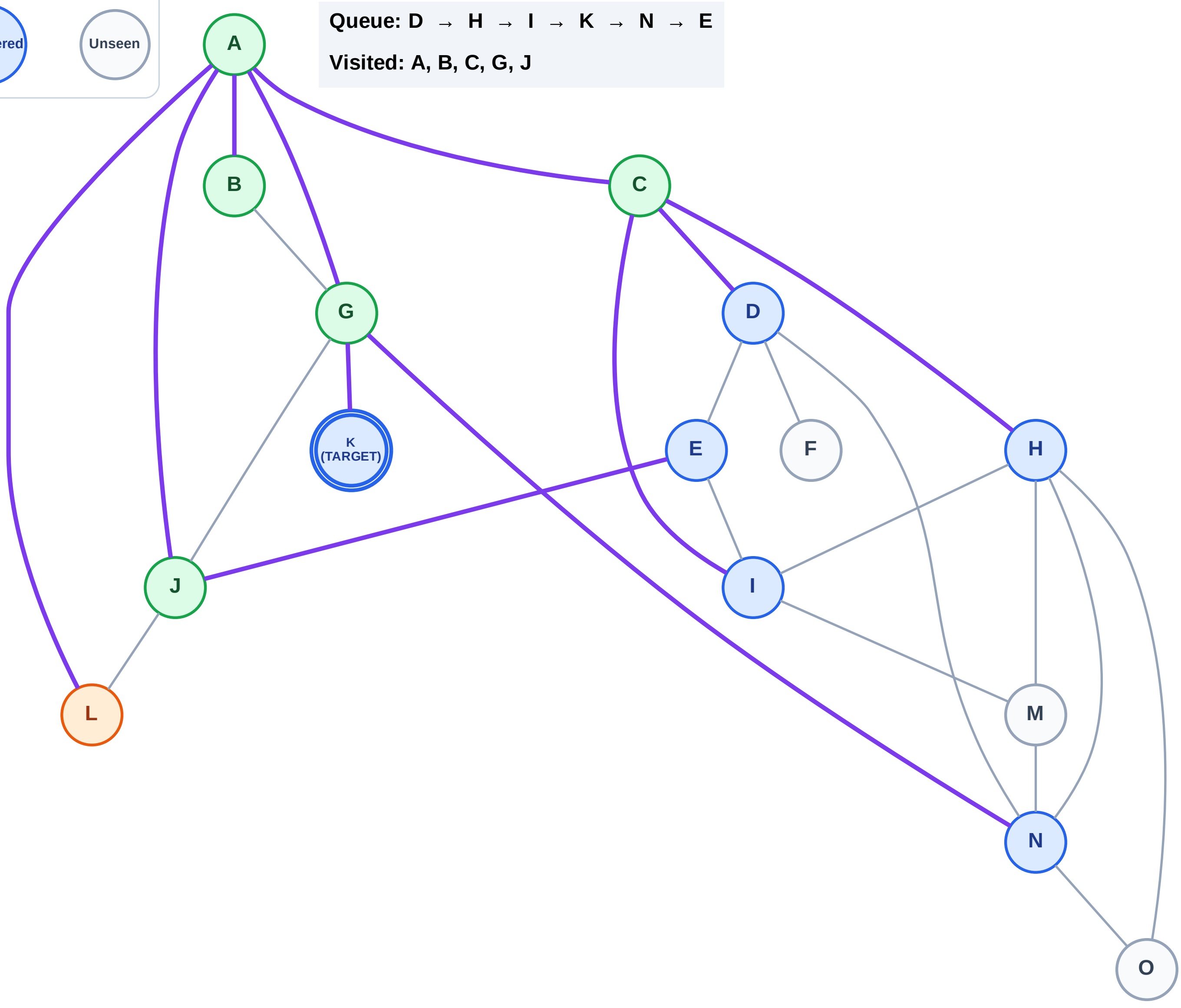
Processing

Discovered

Unseen

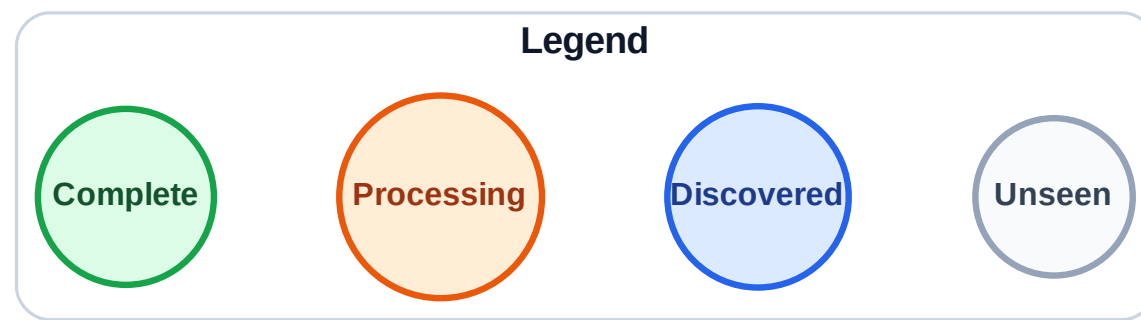
Queue: D → H → I → K → N → E

Visited: A, B, C, G, J



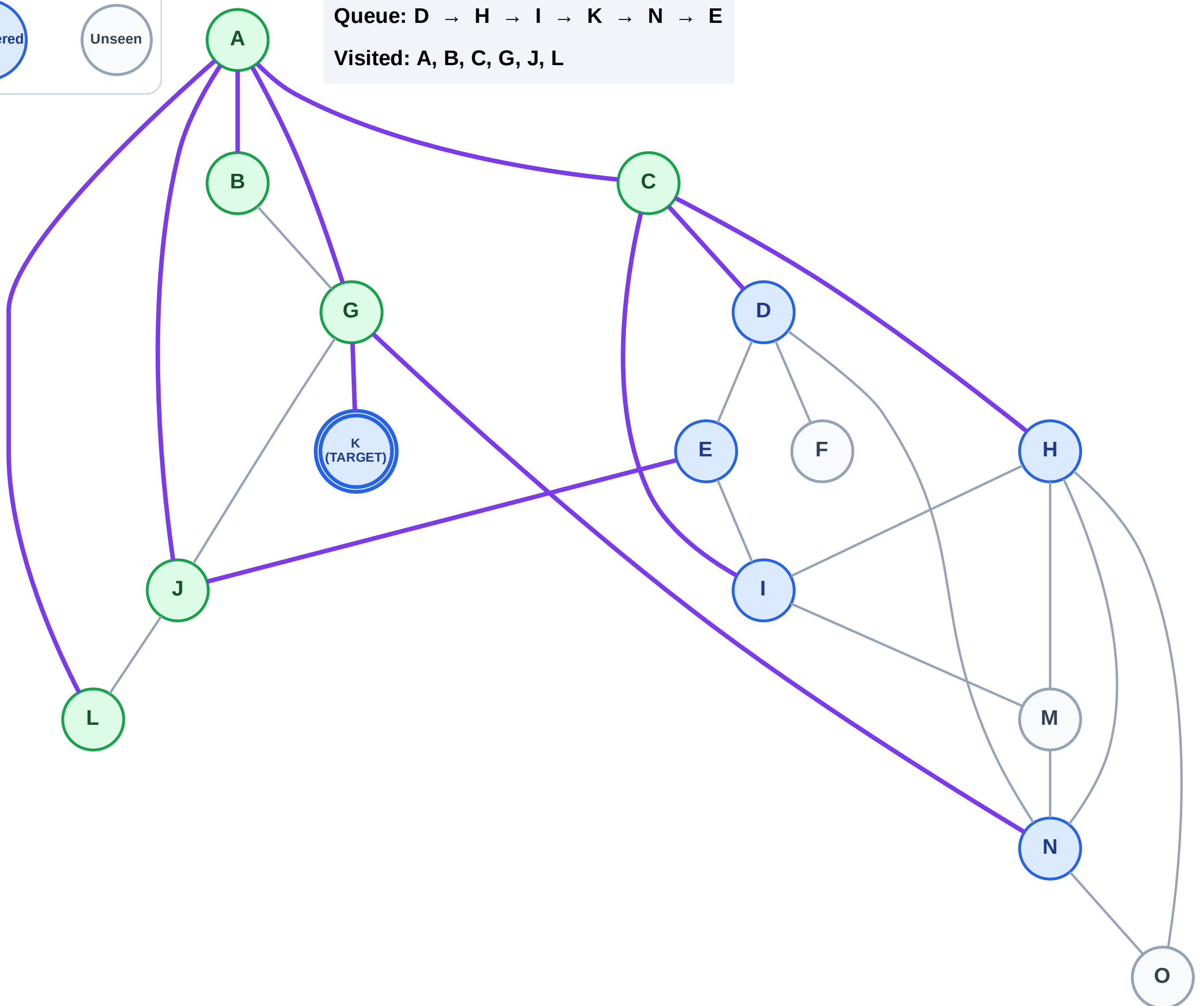
BFS Traversal

Step 13: No new neighbors from L



Queue: D → H → I → K → N → E

Visited: A, B, C, G, J, L



BFS Traversal

Step 14: Process node D

Legend

Complete

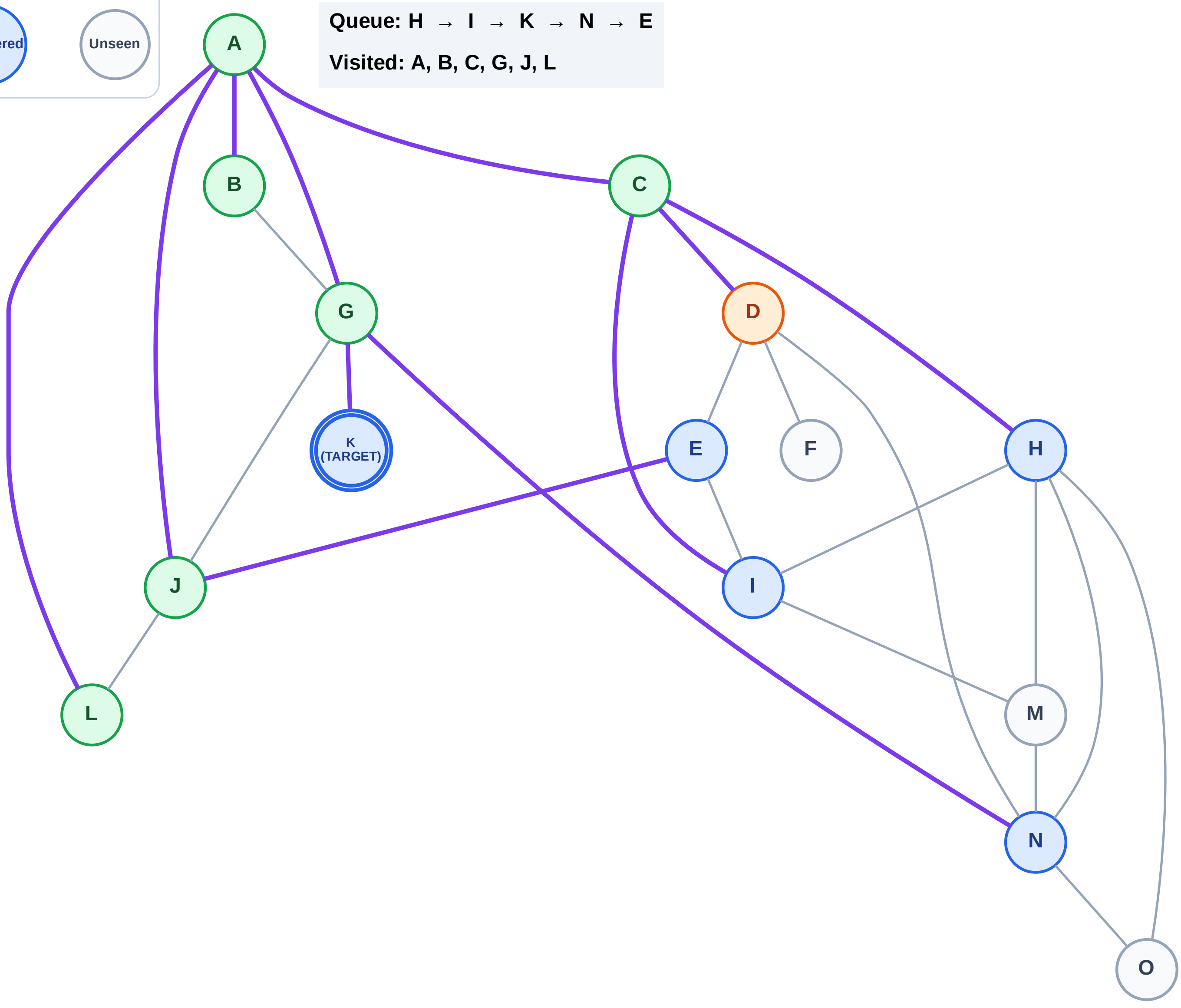
Processing

Discovered

Unseen

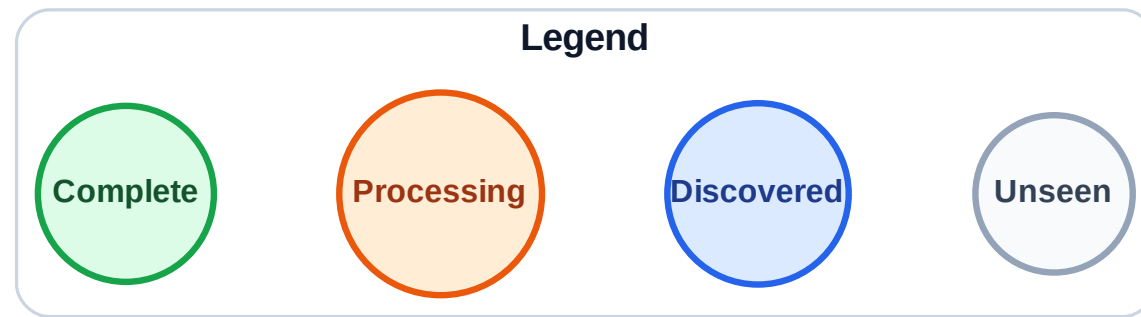
Queue: H → I → K → N → E

Visited: A, B, C, G, J, L



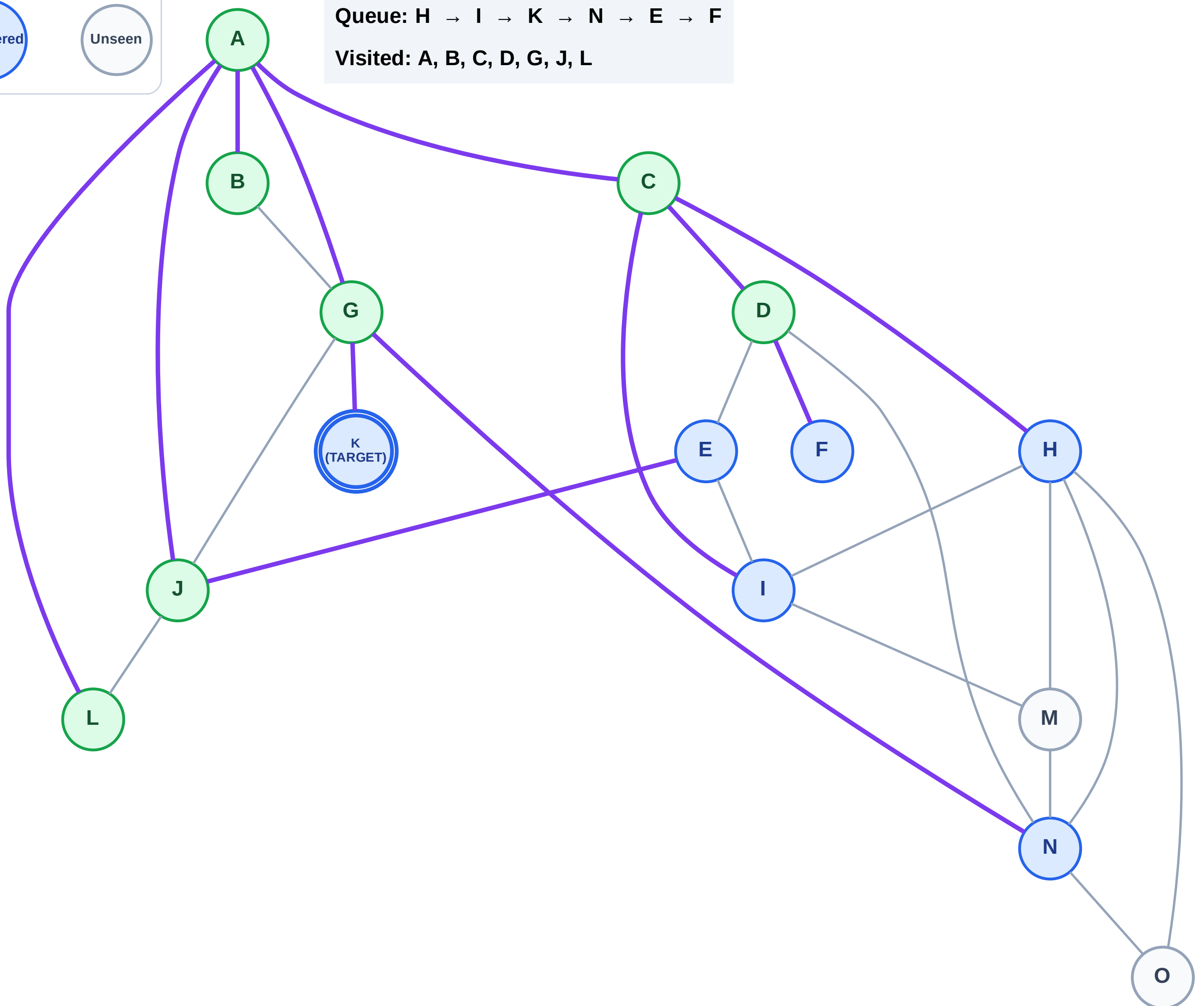
BFS Traversal

Step 15: Discover F from D



Queue: H → I → K → N → E → F

Visited: A, B, C, D, G, J, L



BFS Traversal

Step 16: Process node H

Legend

Complete

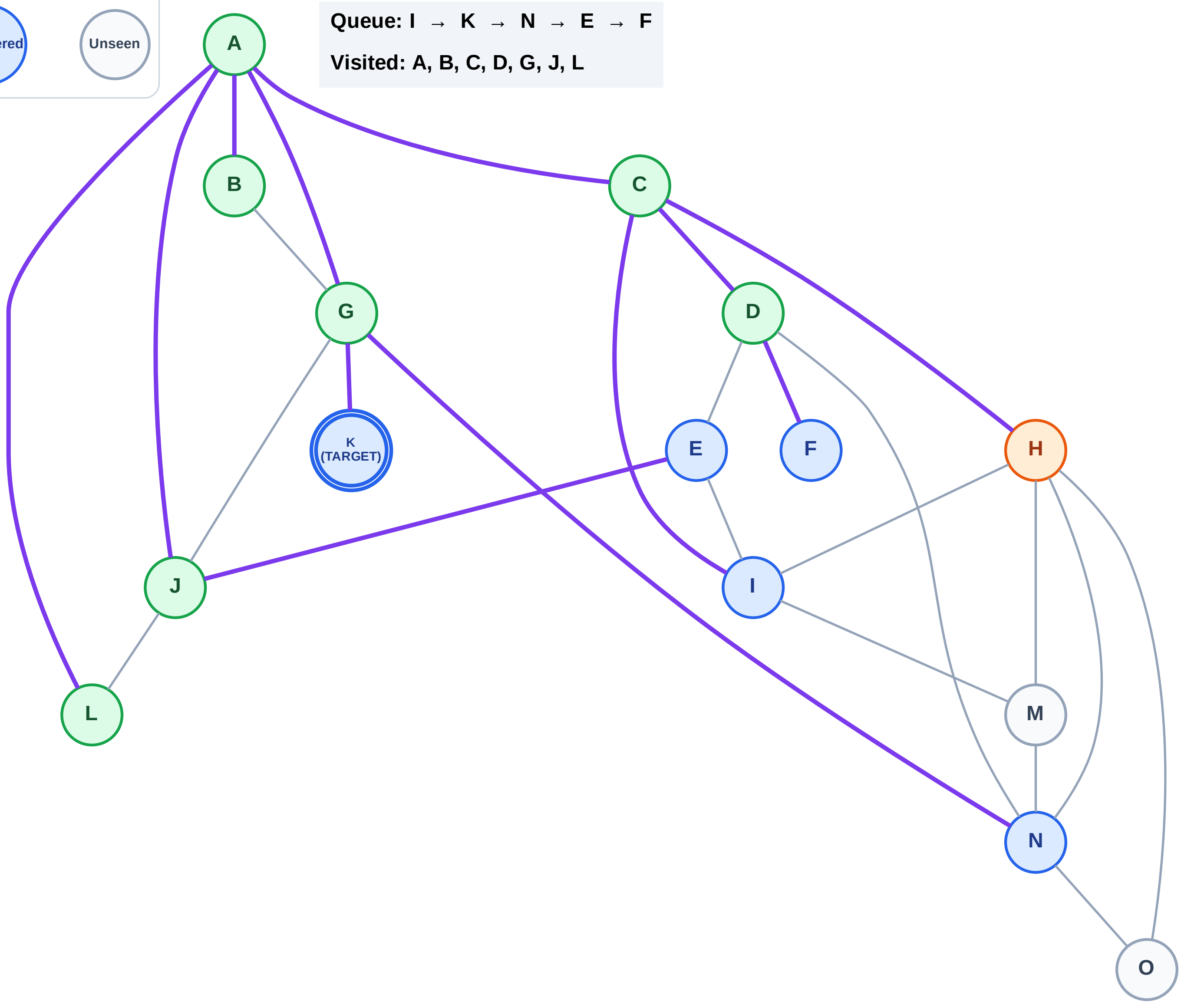
Processing

Discovered

Unseen

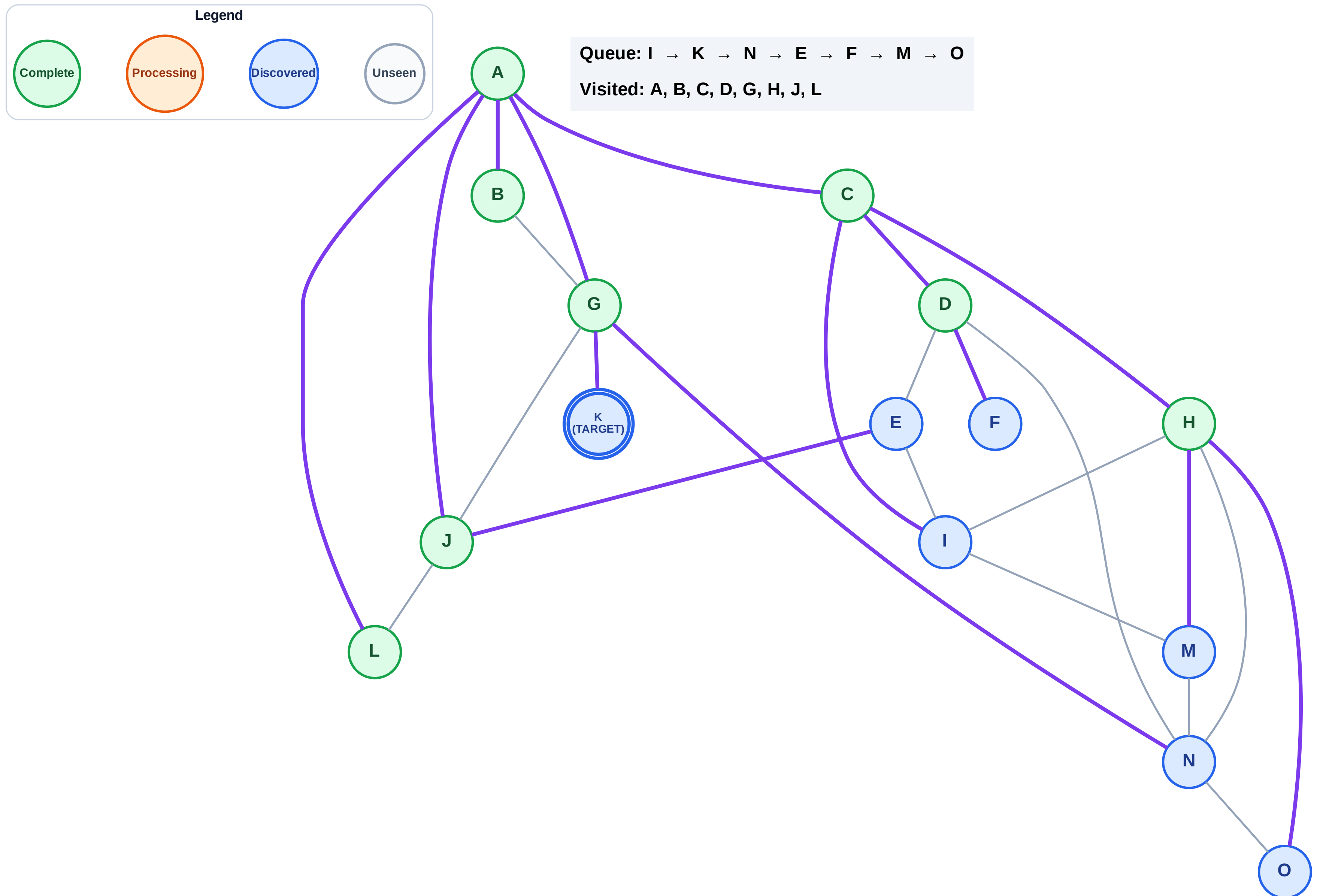
Queue: I → K → N → E → F

Visited: A, B, C, D, G, J, L



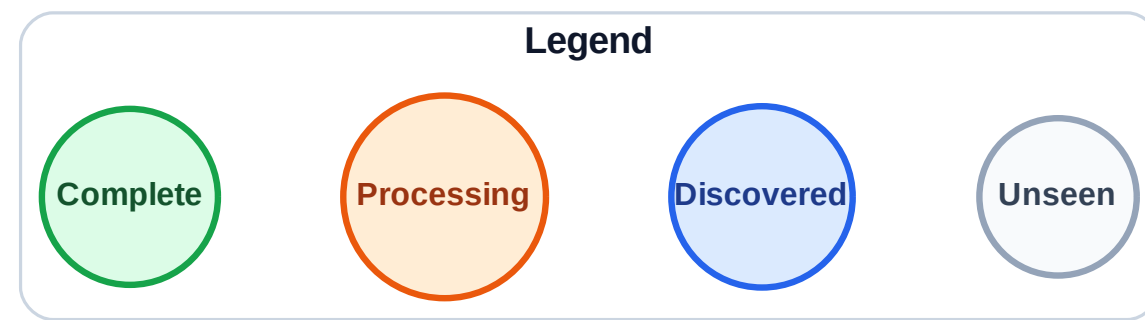
Step 17: Discover M, O from H

Step 17: Discover M, O from H



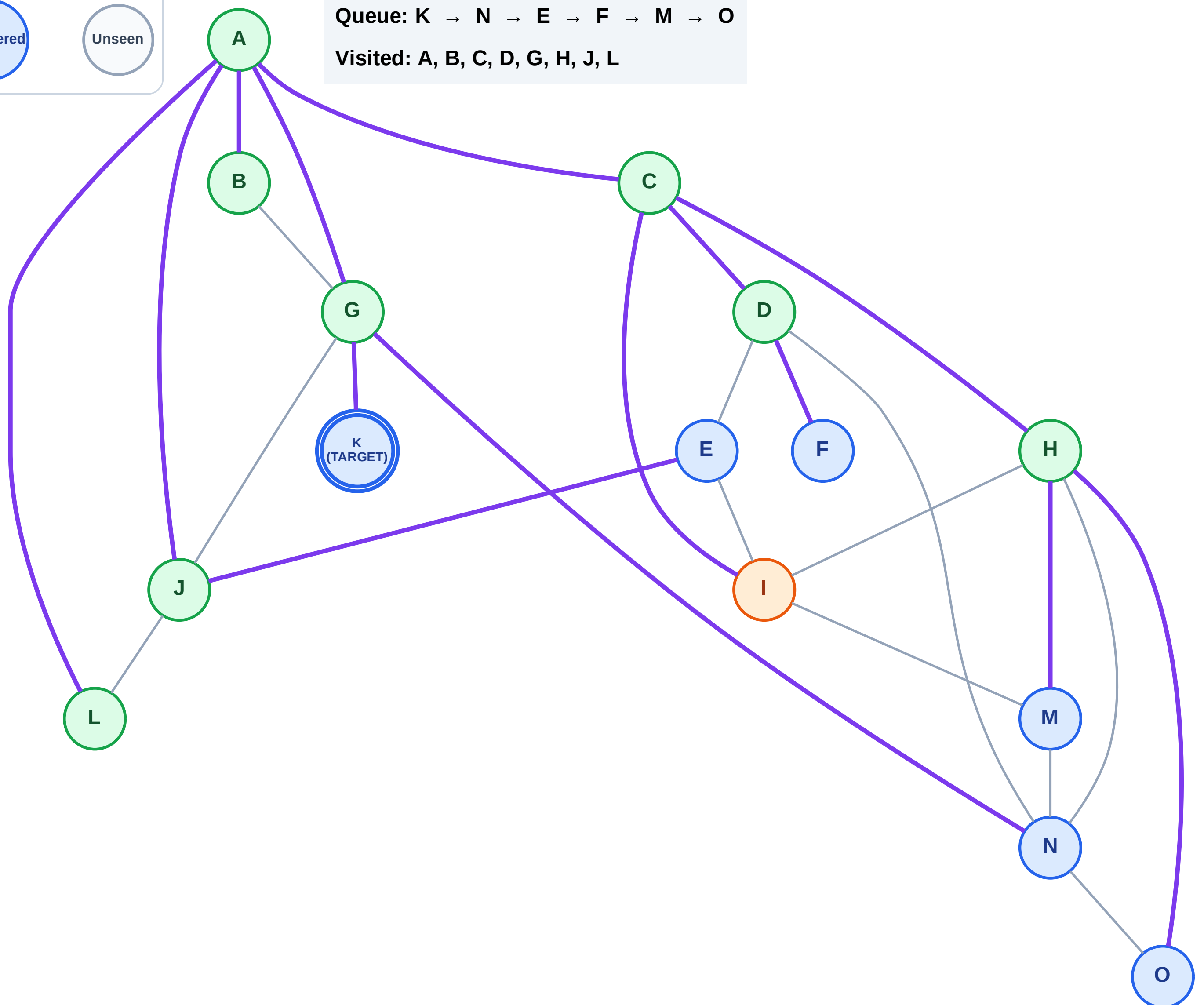
BFS Traversal

Step 18: Process node I



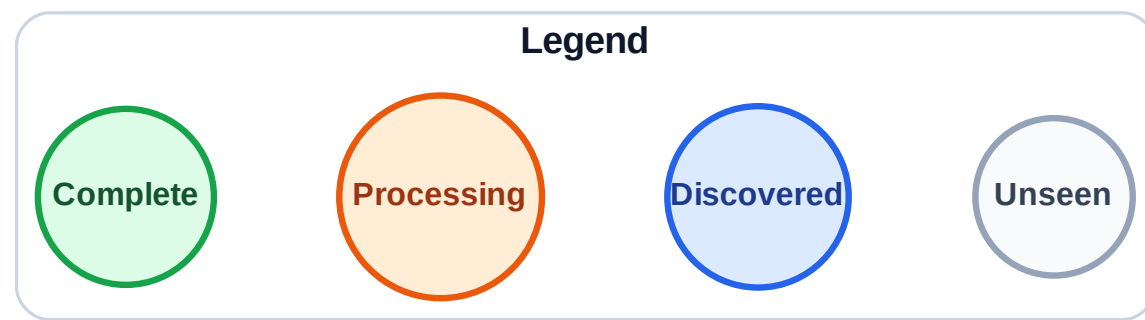
Queue: K → N → E → F → M → O

Visited: A, B, C, D, G, H, J, L



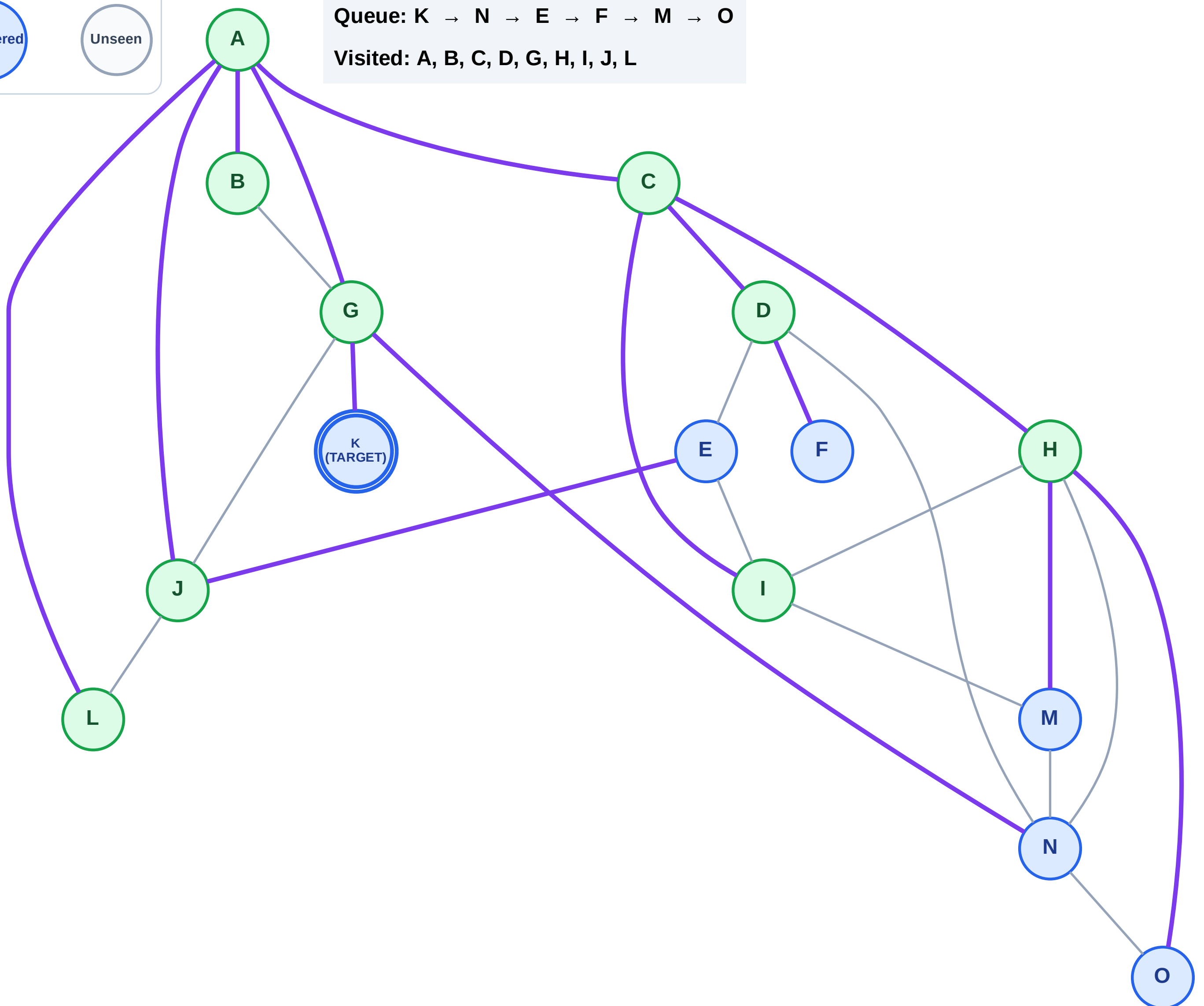
BFS Traversal

Step 19: No new neighbors from I



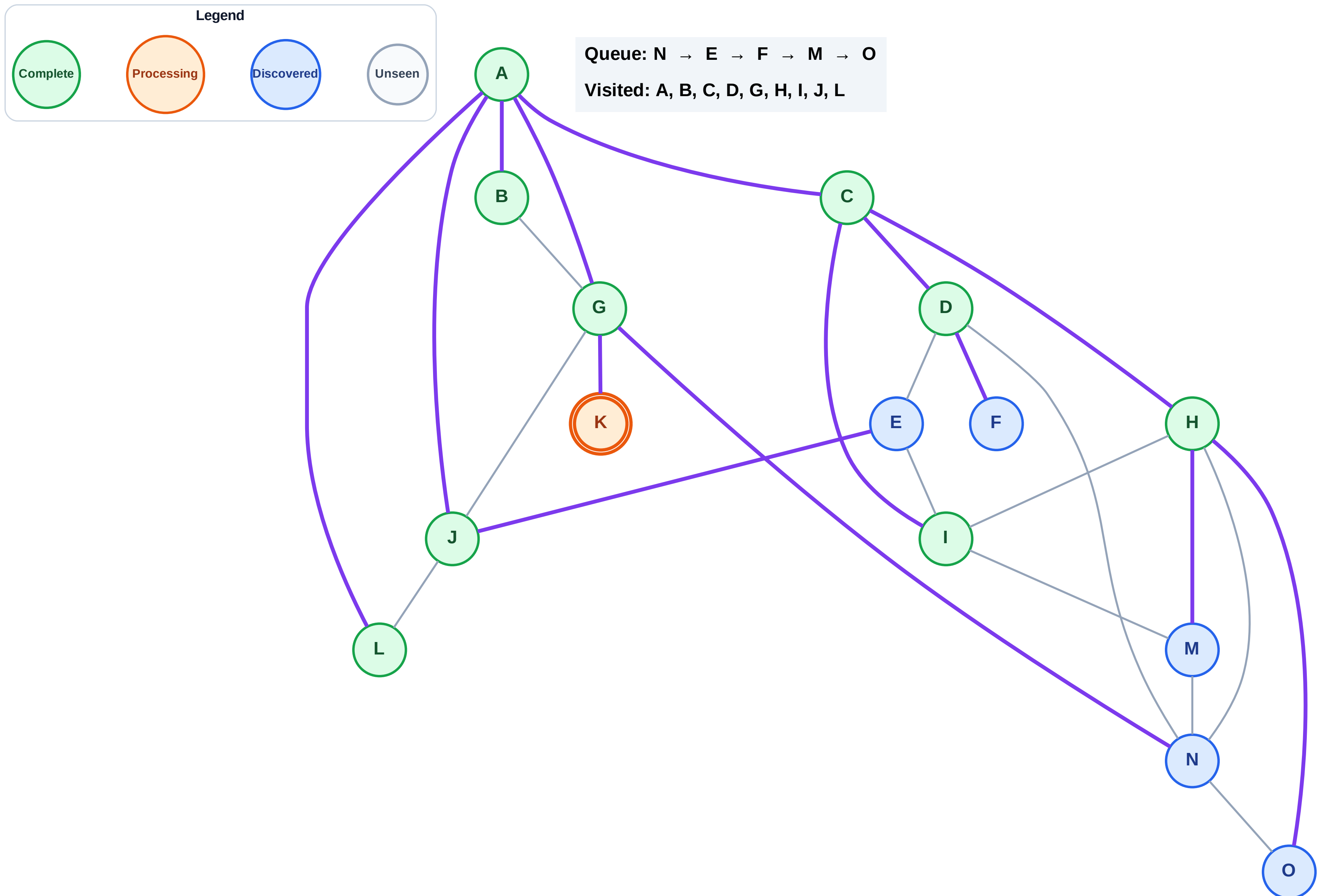
Queue: K → N → E → F → M → O

Visited: A, B, C, D, G, H, I, J, L



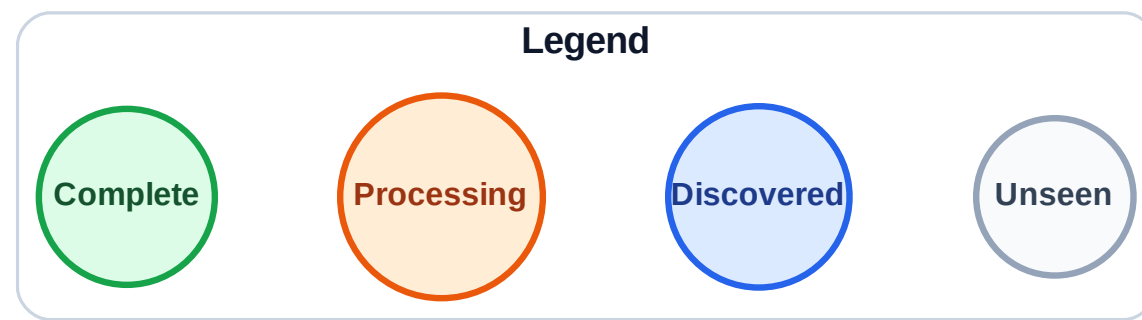
Step 20: Process node K

Step 20: Process node K



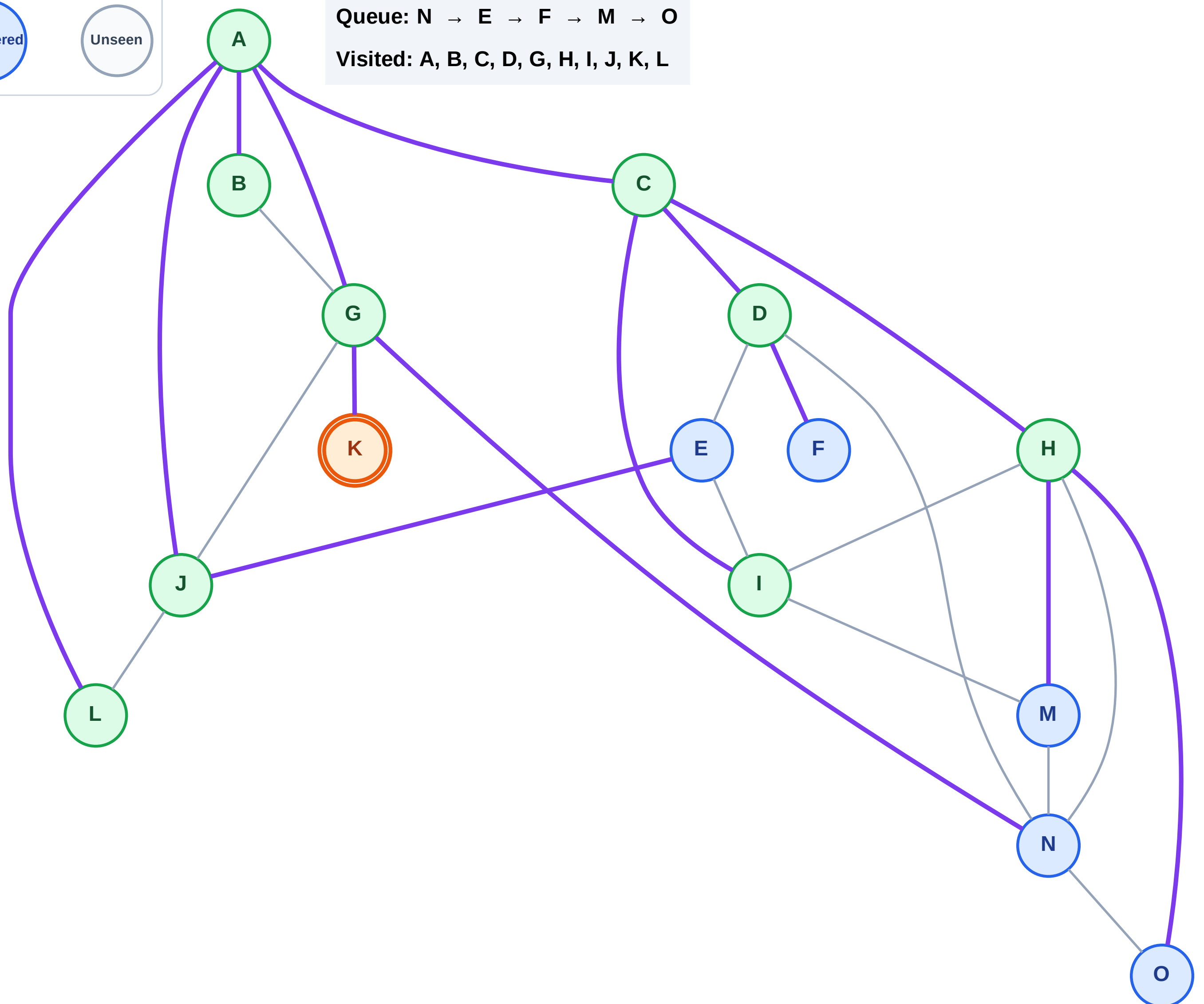
BFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → G → K



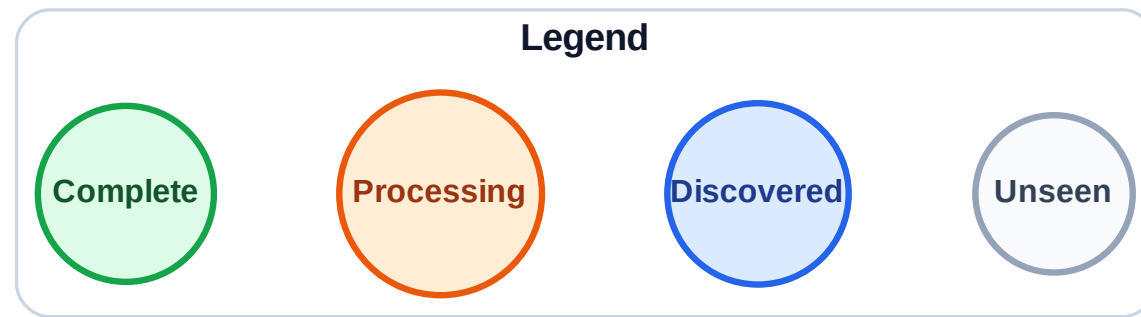
Queue: N → E → F → M → O

Visited: A, B, C, D, G, H, I, J, K, L



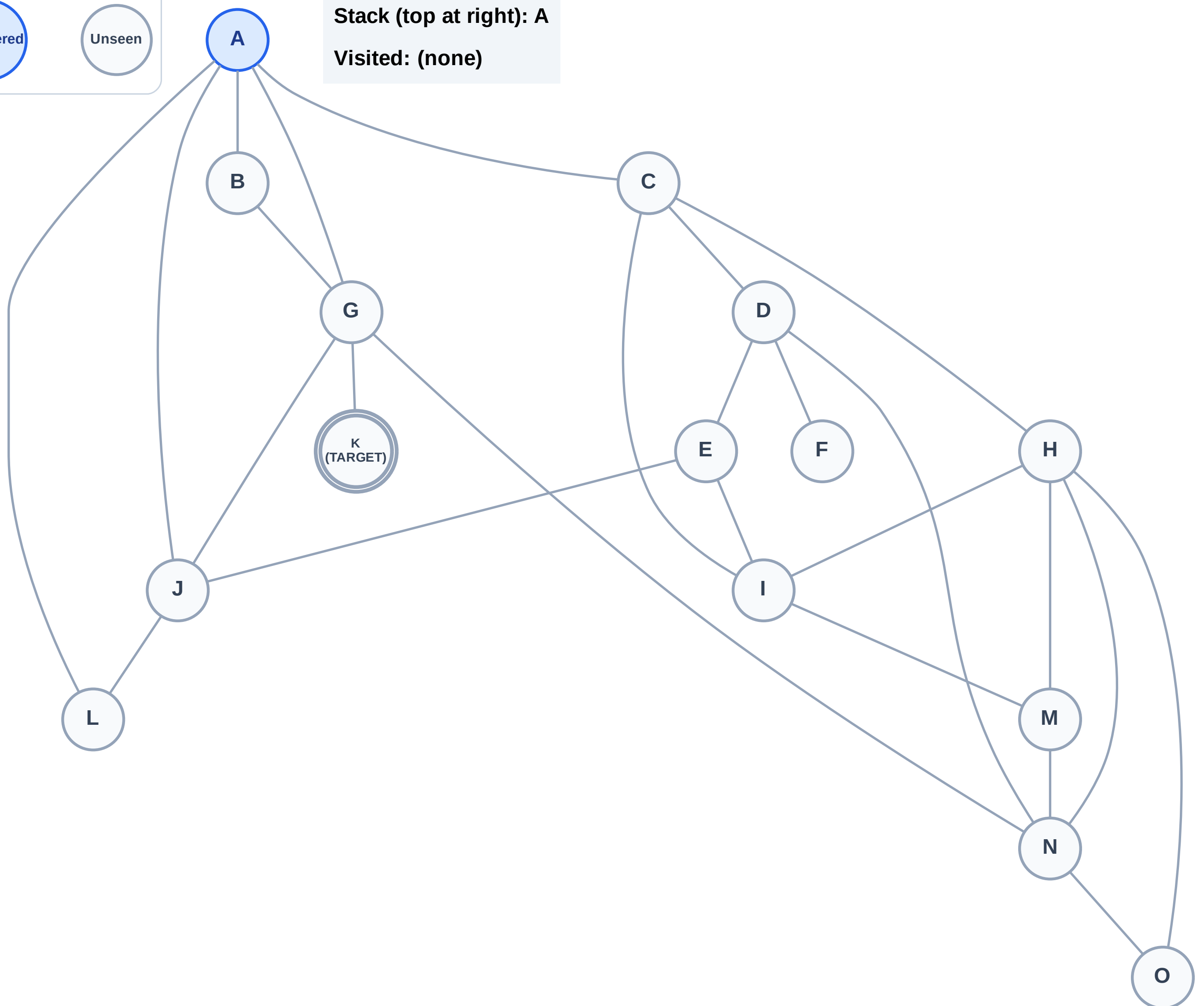
DFS Traversal

Step 1: Start at A (Seeking target node K)



Stack (top at right): A

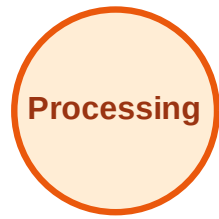
Visited: (none)



DFS Traversal

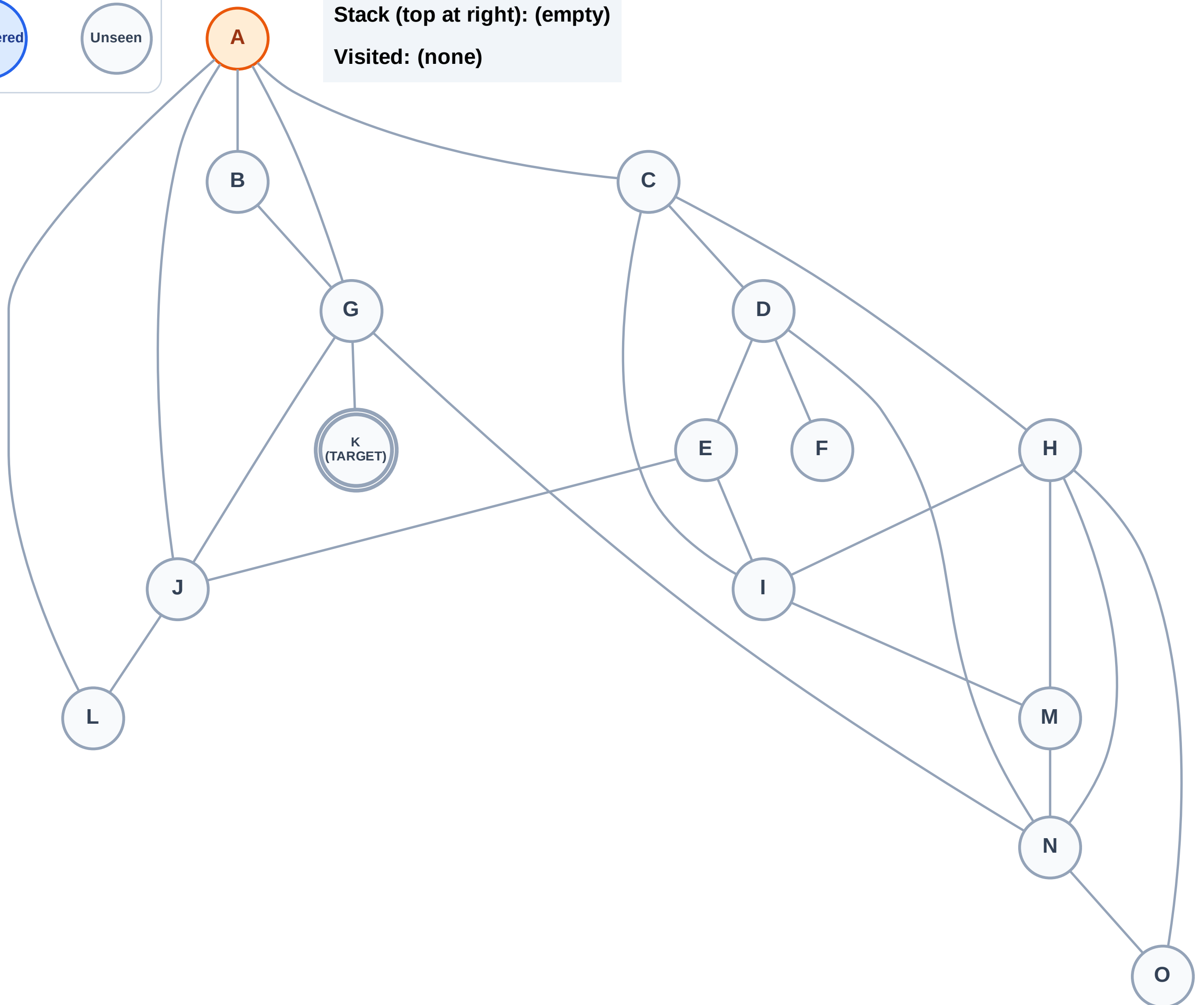
Step 2: Process node A

Legend



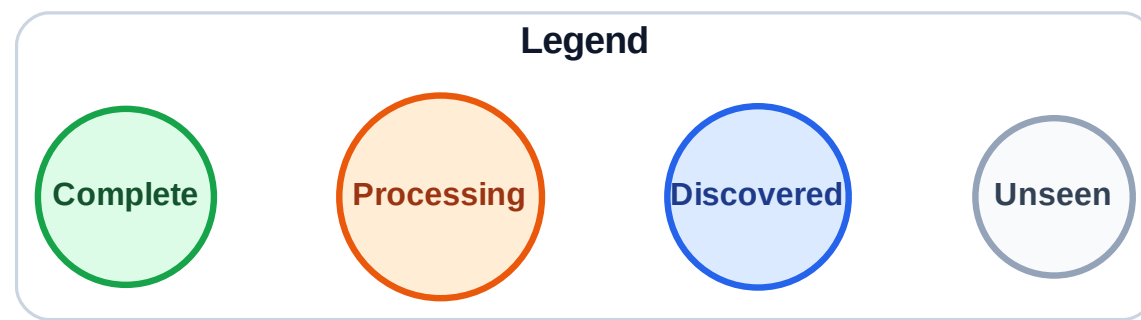
Stack (top at right): (empty)

Visited: (none)



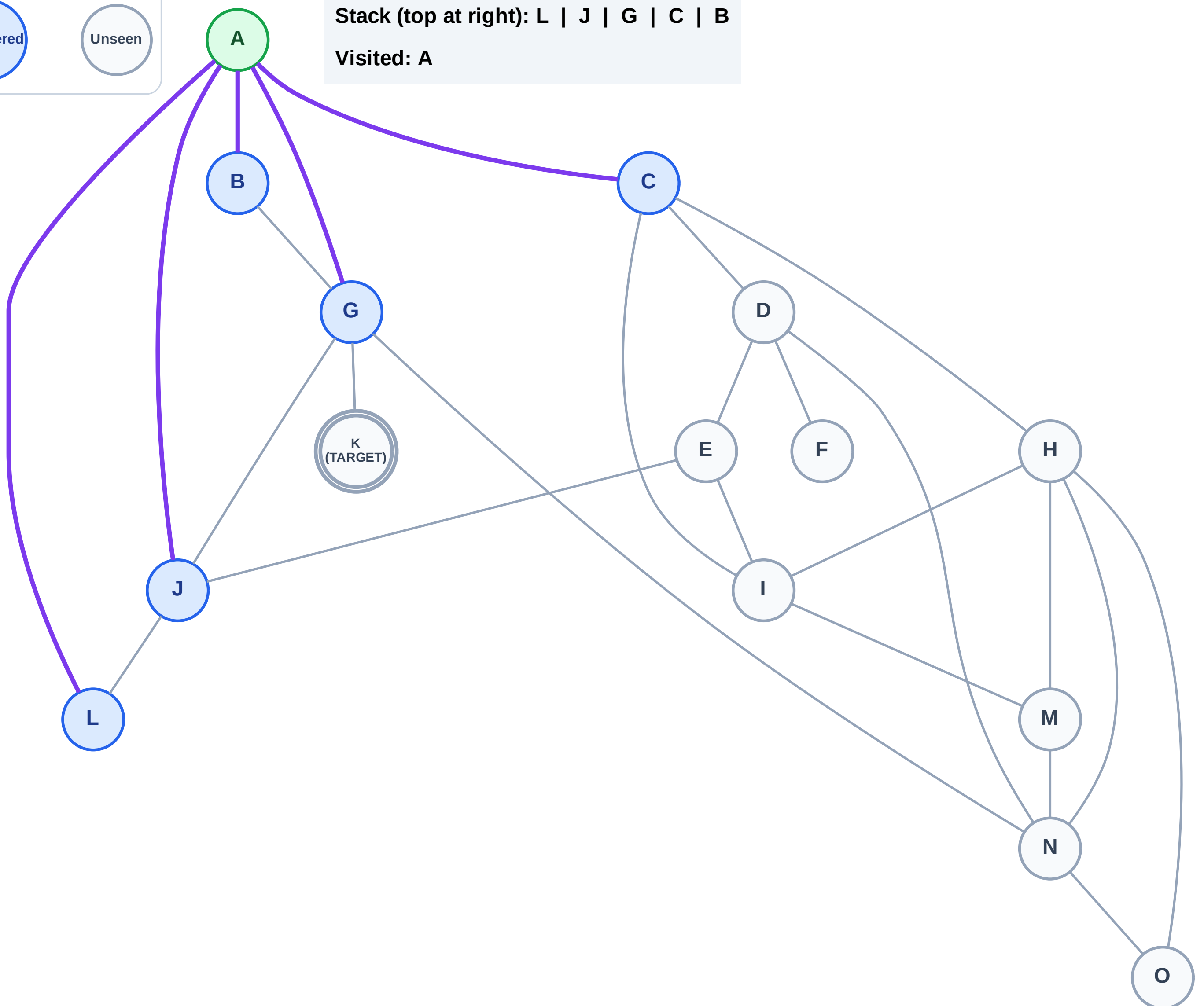
DFS Traversal

Step 3: Discover L, J, G, C, B from A



Stack (top at right): L | J | G | C | B

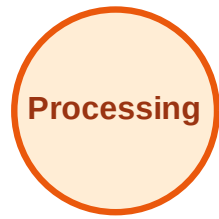
Visited: A



DFS Traversal

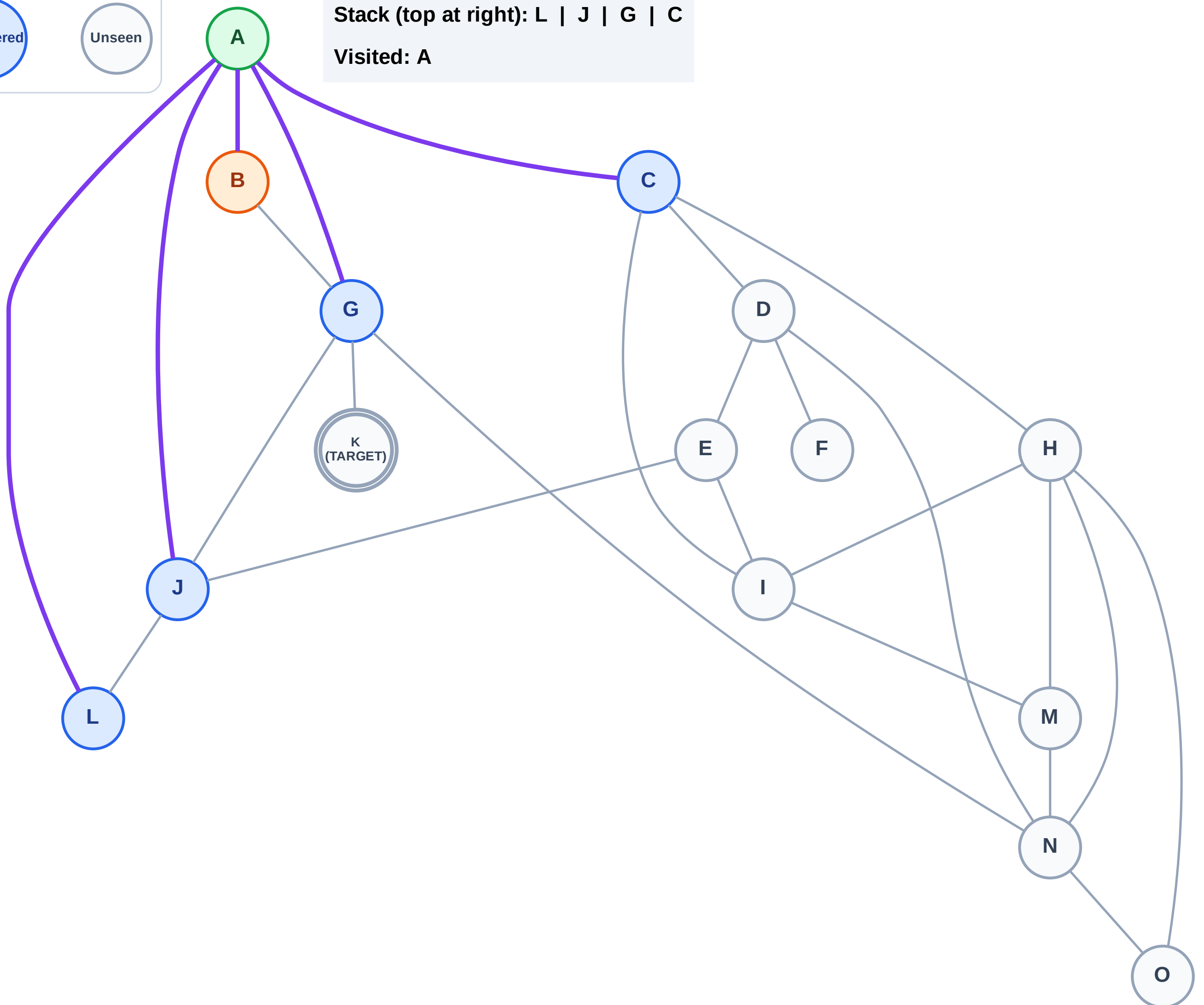
Step 4: Process node B

Legend



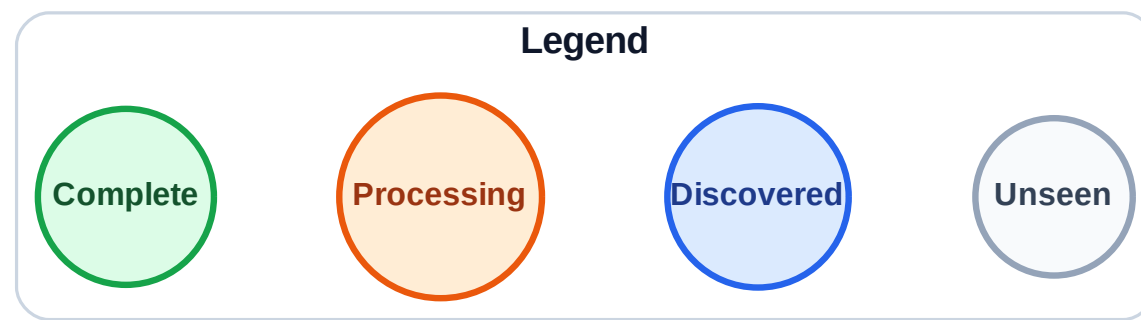
Stack (top at right): L | J | G | C

Visited: A



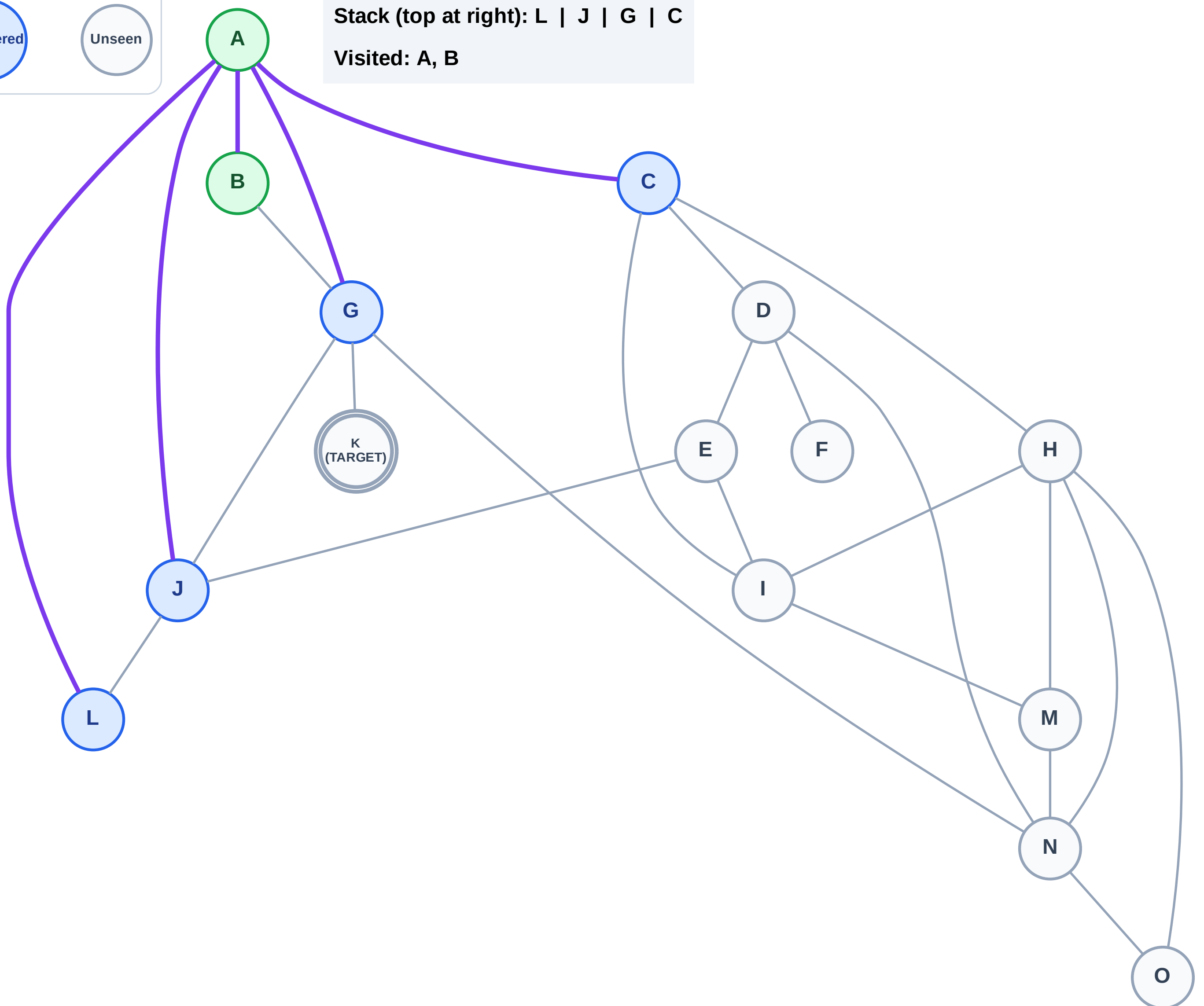
DFS Traversal

Step 5: No new neighbors from B



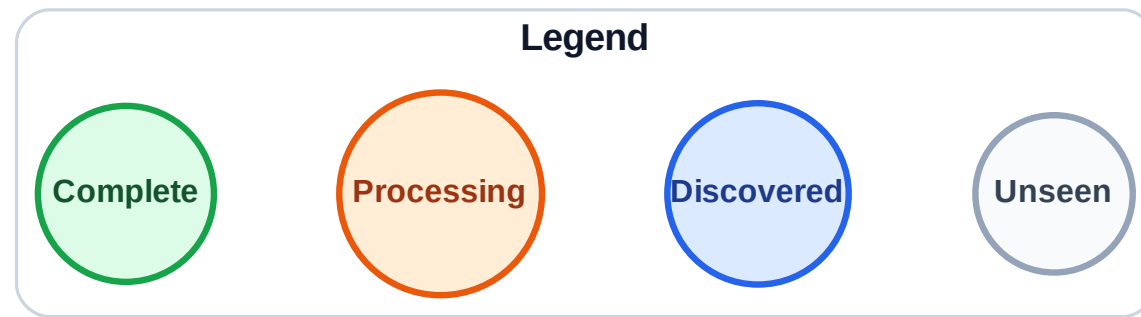
Stack (top at right): L | J | G | C

Visited: A, B



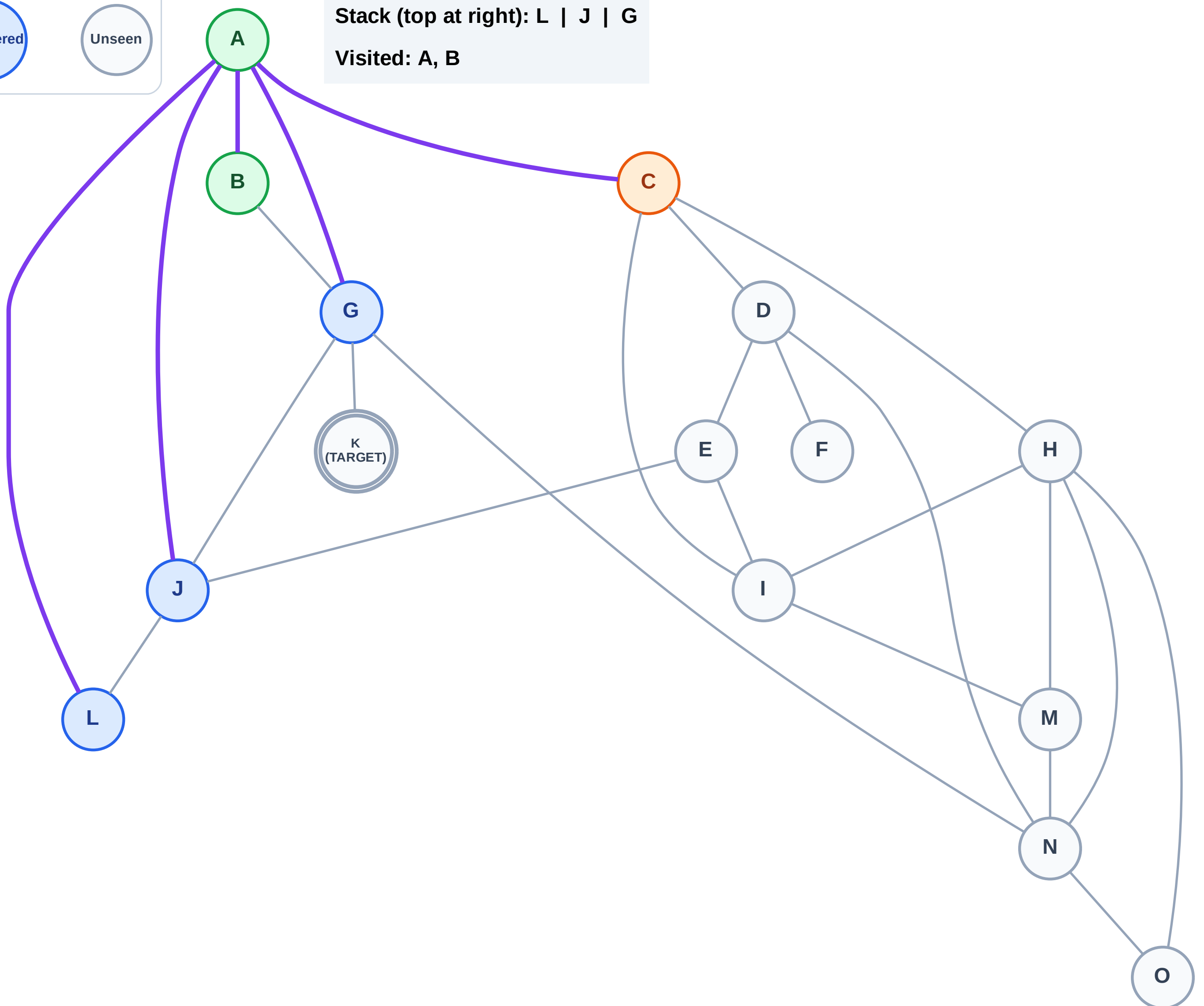
DFS Traversal

Step 6: Process node C



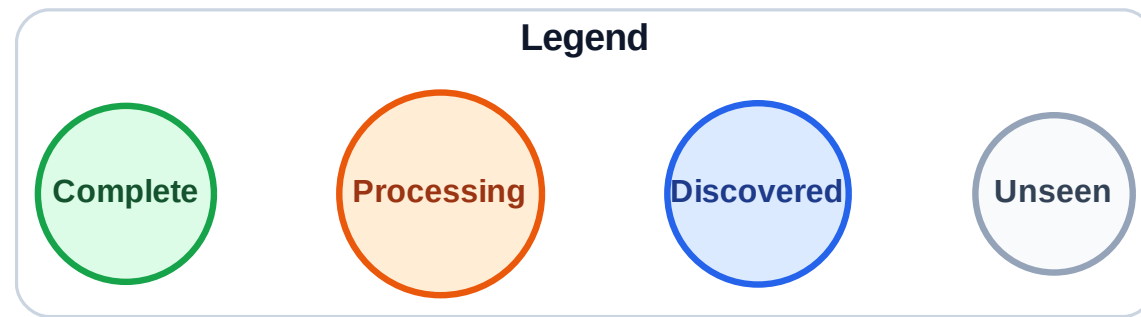
Stack (top at right): L | J | G

Visited: A, B



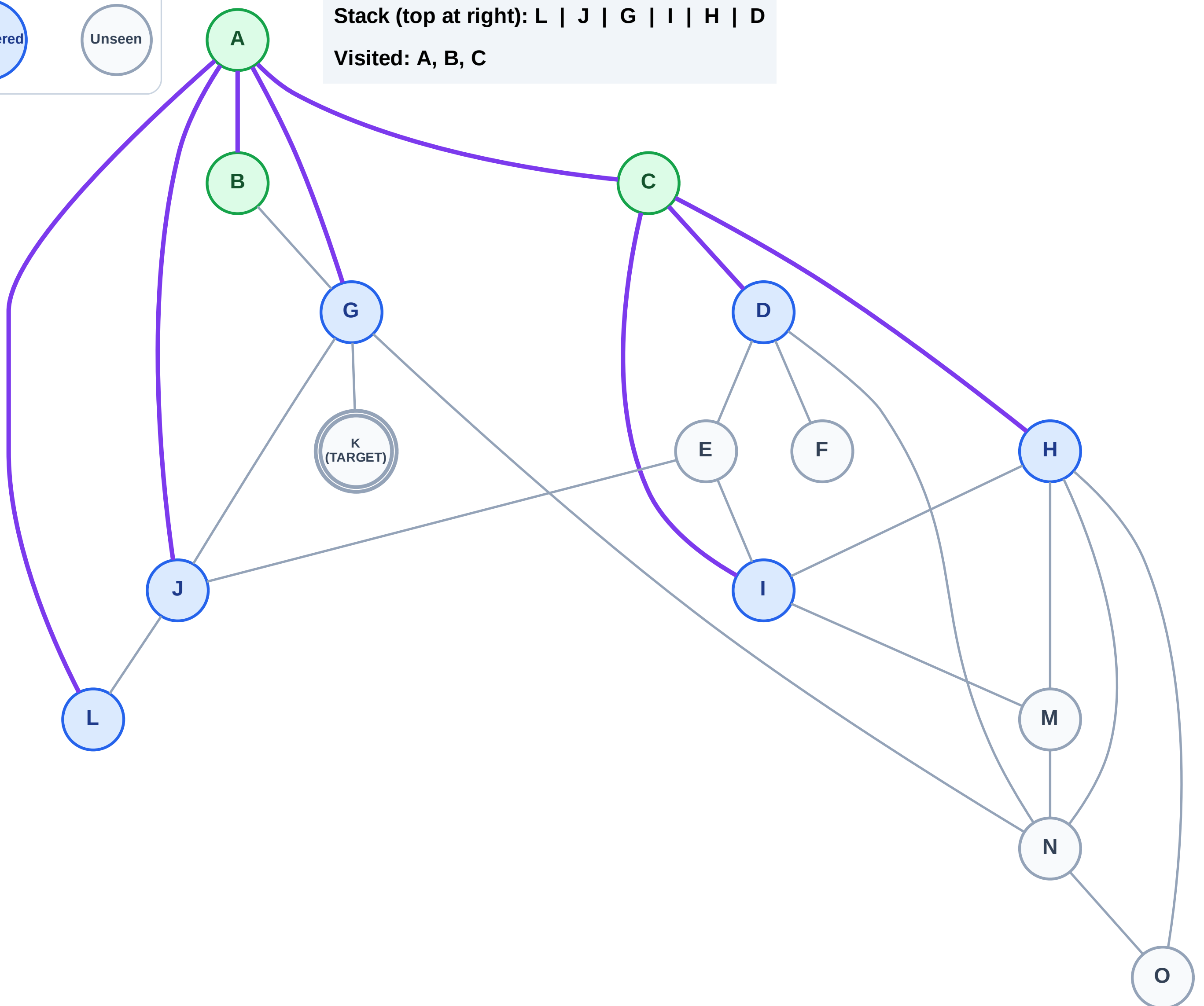
DFS Traversal

Step 7: Discover I, H, D from C



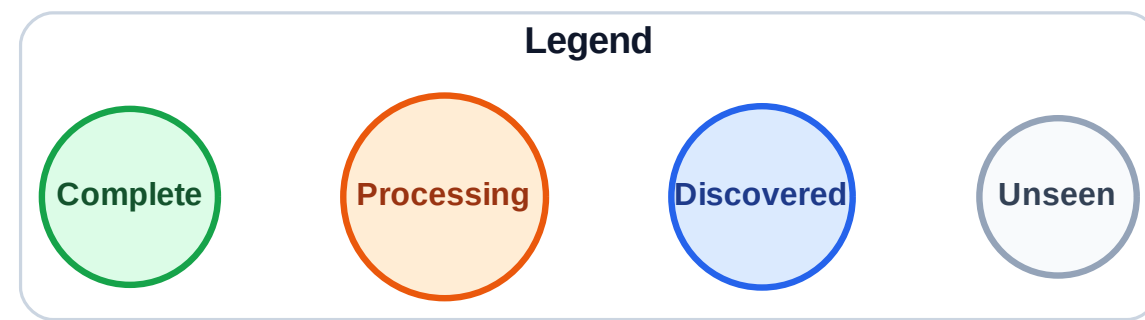
Stack (top at right): L | J | G | I | H | D

Visited: A, B, C



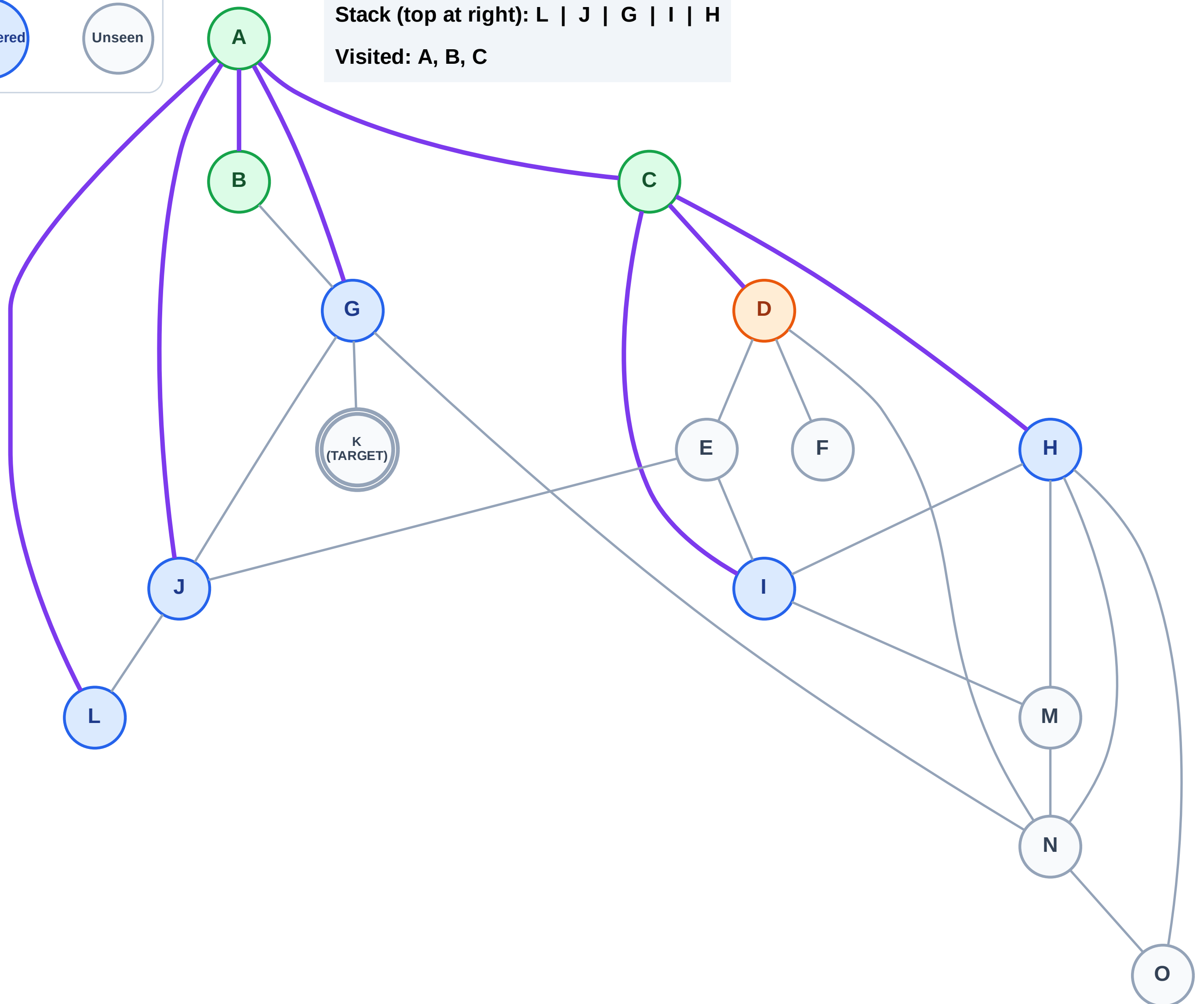
DFS Traversal

Step 8: Process node D



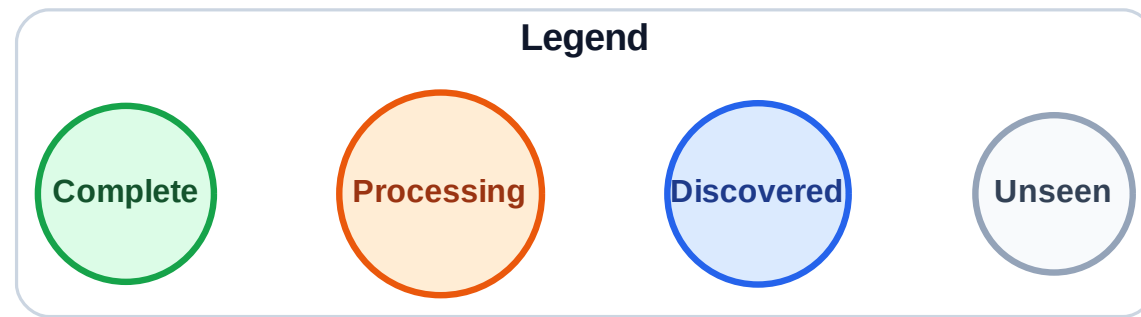
Stack (top at right): L | J | G | I | H

Visited: A, B, C



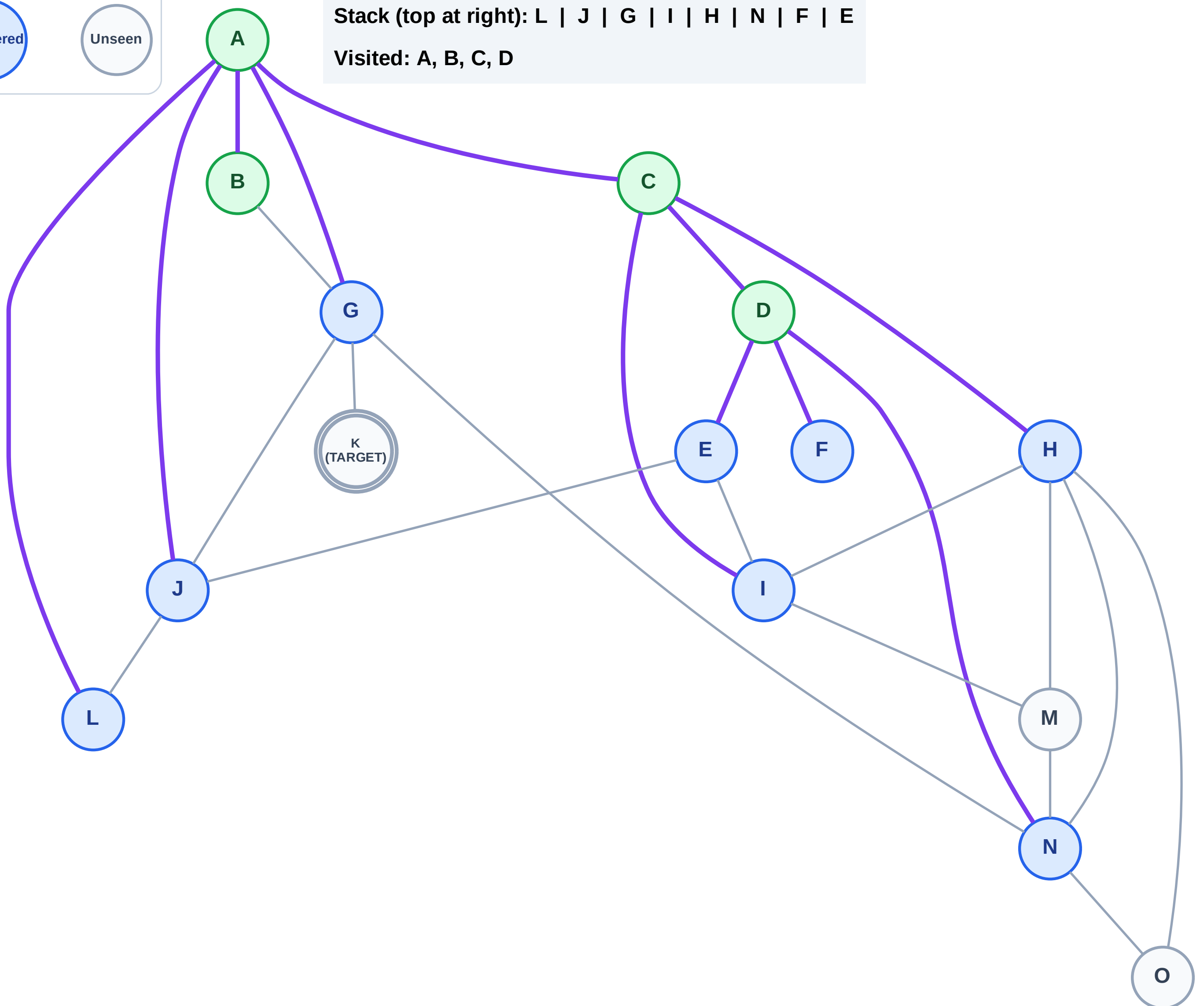
DFS Traversal

Step 9: Discover N, F, E from D



Stack (top at right): L | J | G | I | H | N | F | E

Visited: A, B, C, D



DFS Traversal

Step 10: Process node E

Legend

Complete

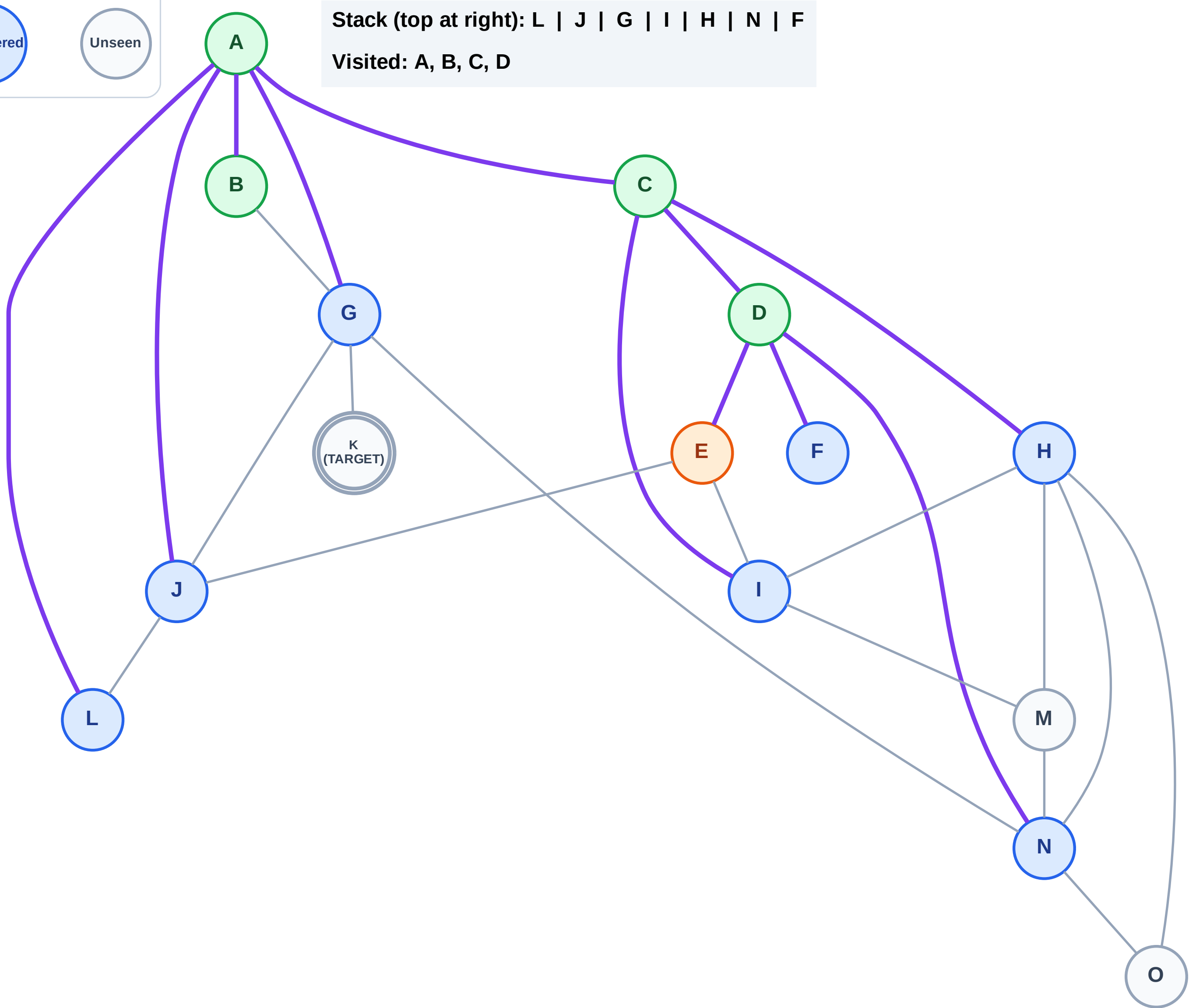
Processing

Discovered

Unseen

Stack (top at right): L | J | G | I | H | N | F

Visited: A, B, C, D



DFS Traversal

Step 11: No new neighbors from E

Legend

Complete

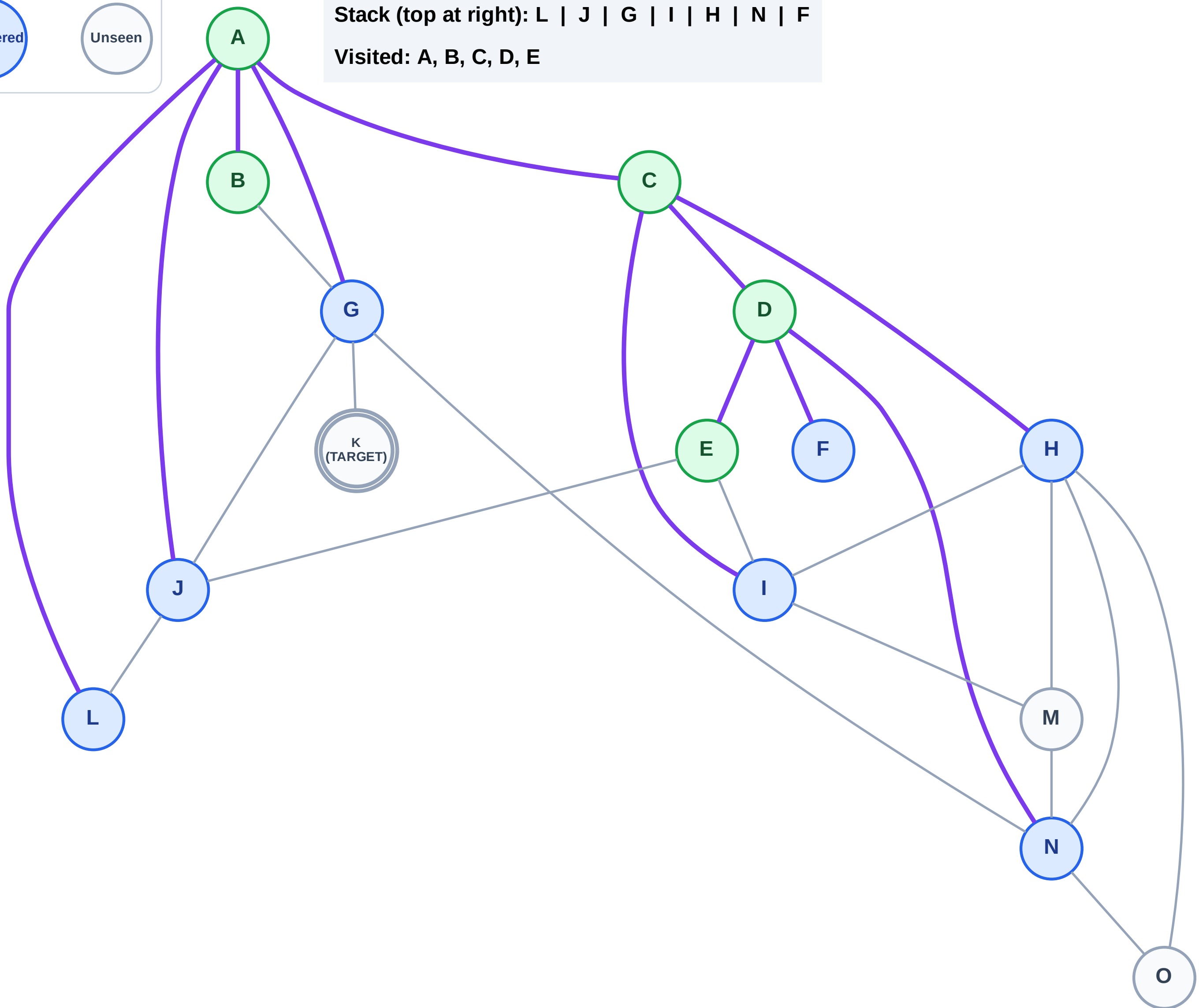
Processing

Discovered

Unseen

Stack (top at right): L | J | G | I | H | N | F

Visited: A, B, C, D, E



DFS Traversal

Step 12: Process node F

Legend

Complete

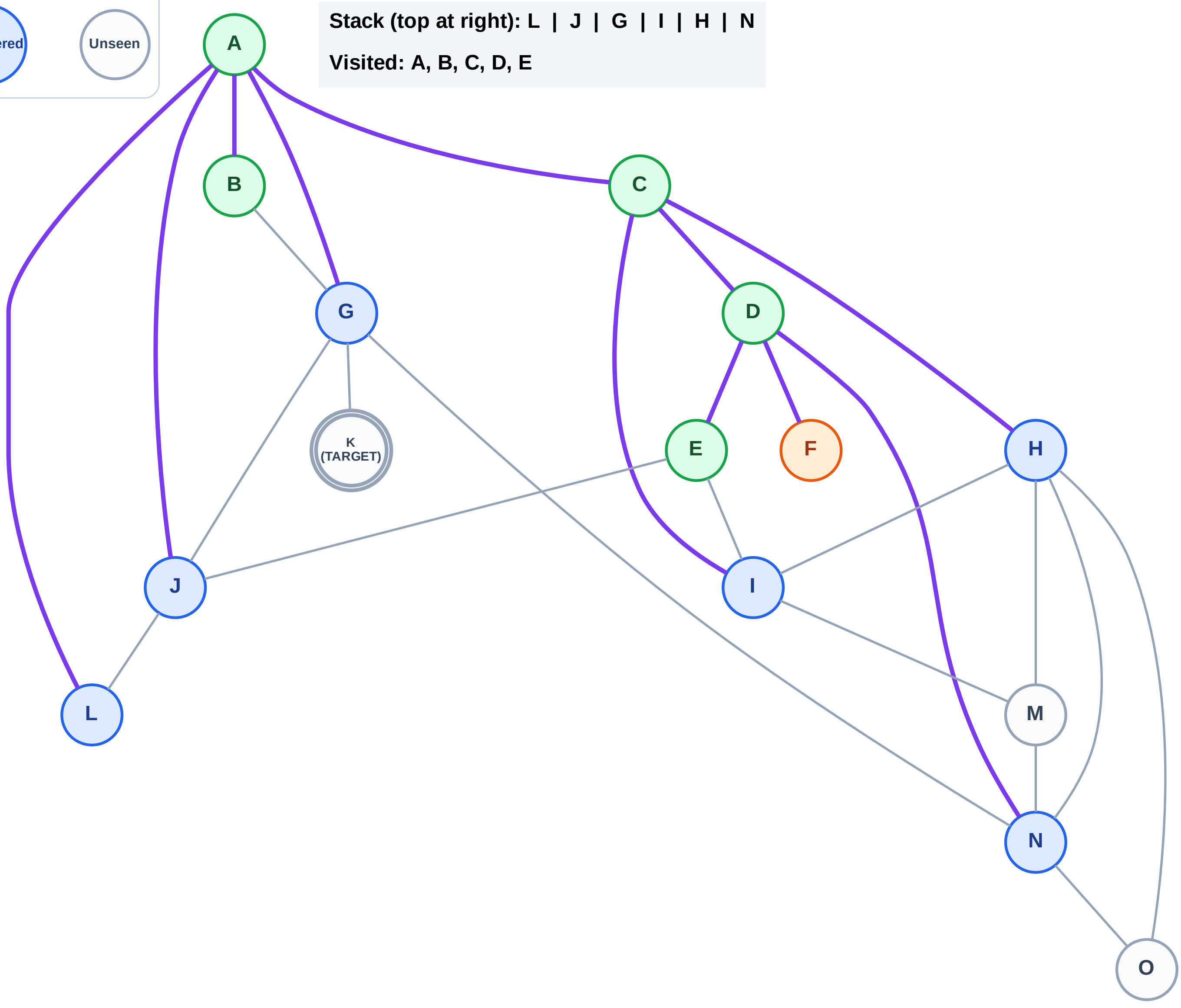
Processing

Discovered

Unseen

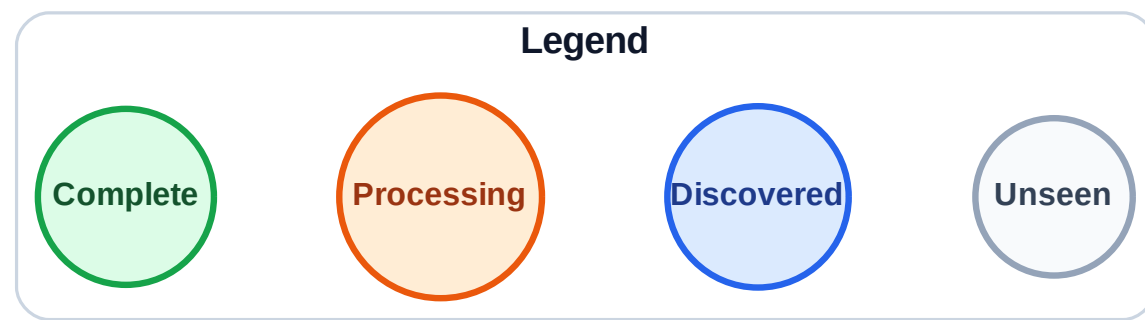
Stack (top at right): L | J | G | I | H | N

Visited: A, B, C, D, E



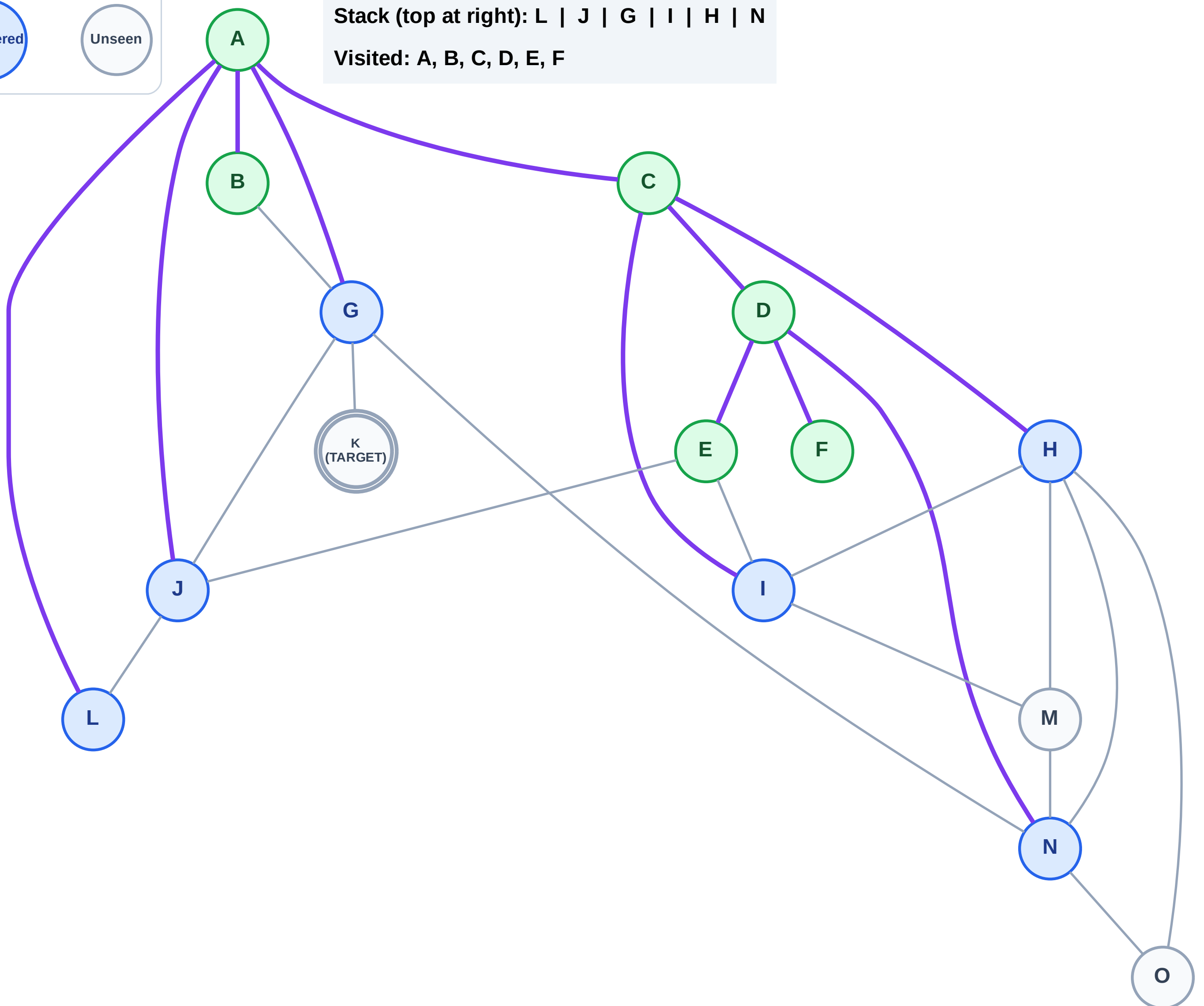
DFS Traversal

Step 13: No new neighbors from F



Stack (top at right): L | J | G | I | H | N

Visited: A, B, C, D, E, F



DFS Traversal

Step 14: Process node N

Legend

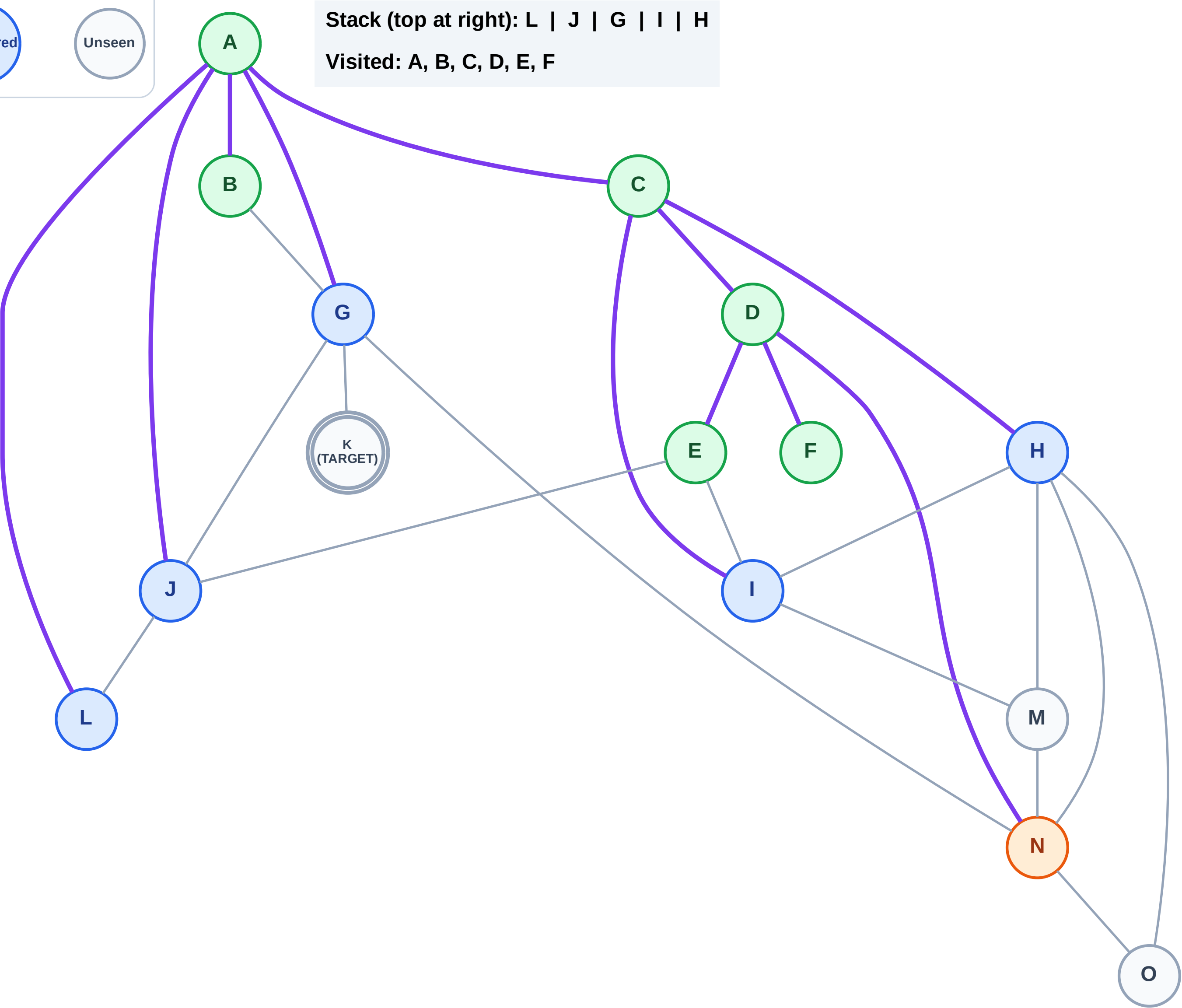
Complete

Processing

Discovered

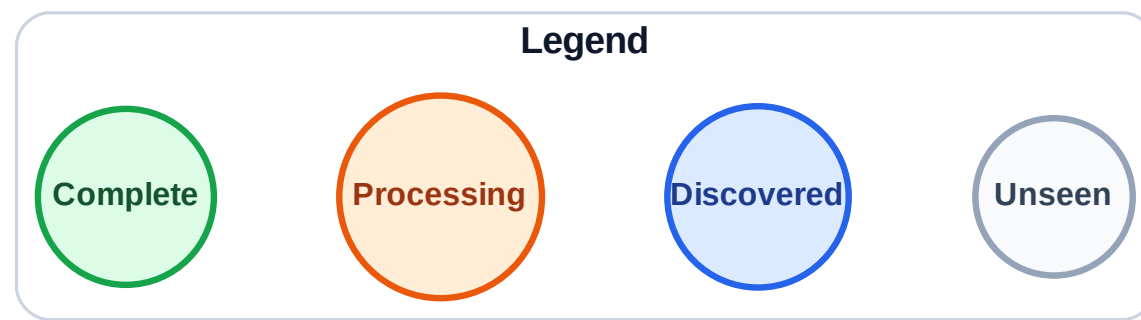
Unseen

Stack (top at right): L | J | G | I | H
Visited: A, B, C, D, E, F



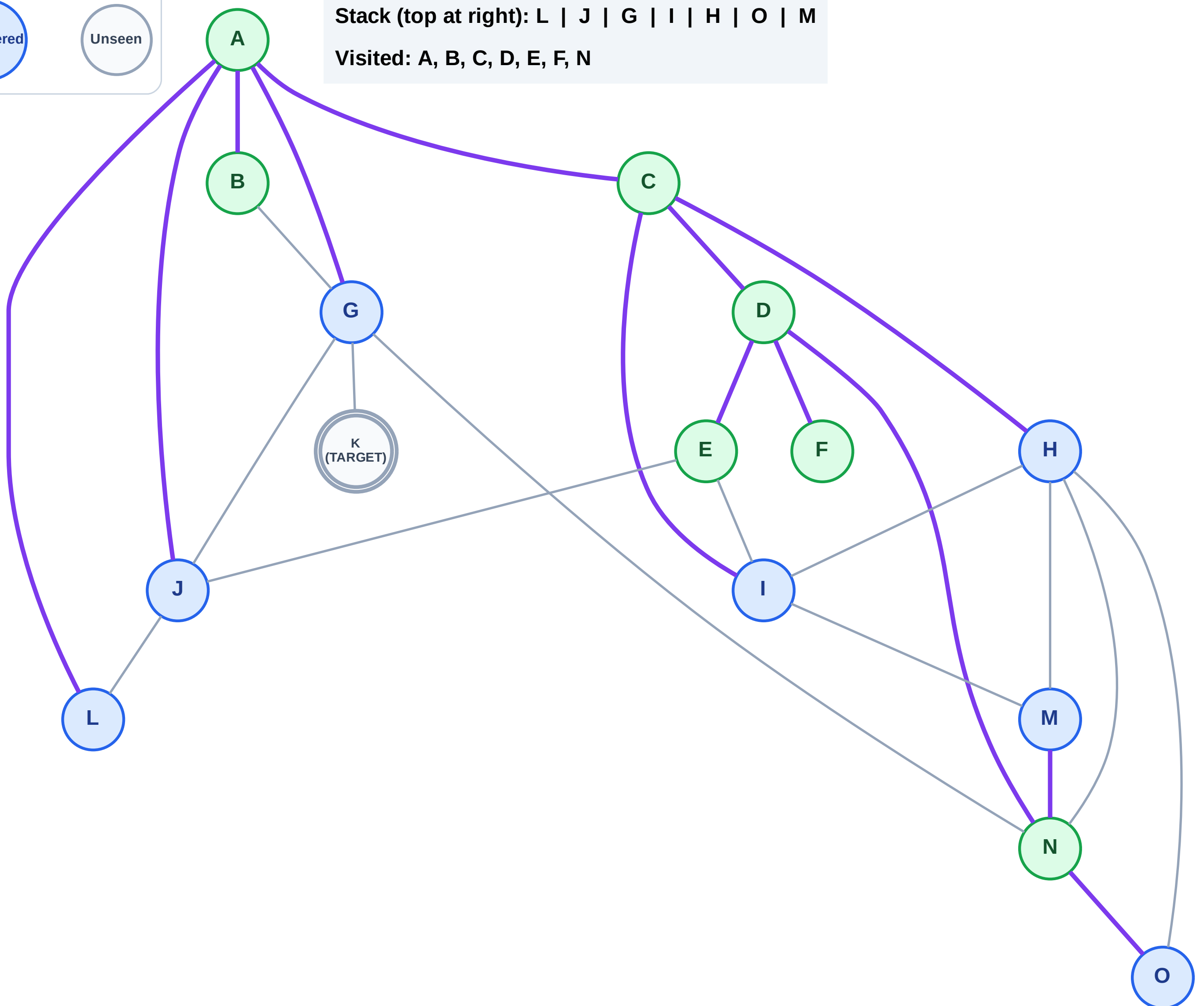
DFS Traversal

Step 15: Discover O, M from N



Stack (top at right): L | J | G | I | H | O | M

Visited: A, B, C, D, E, F, N



DFS Traversal

Step 16: Process node M

Legend

Complete

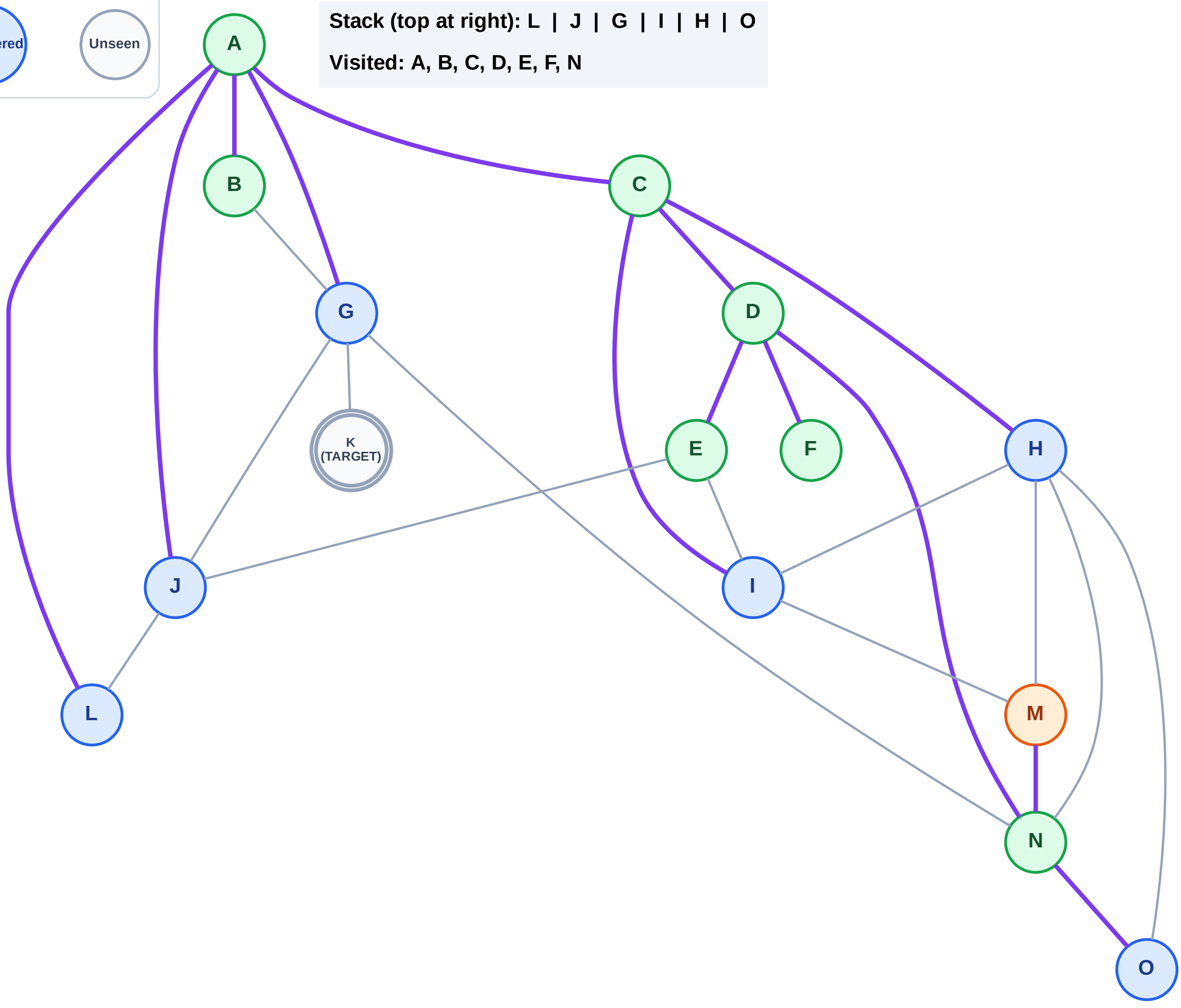
Processing

Discovered

Unseen

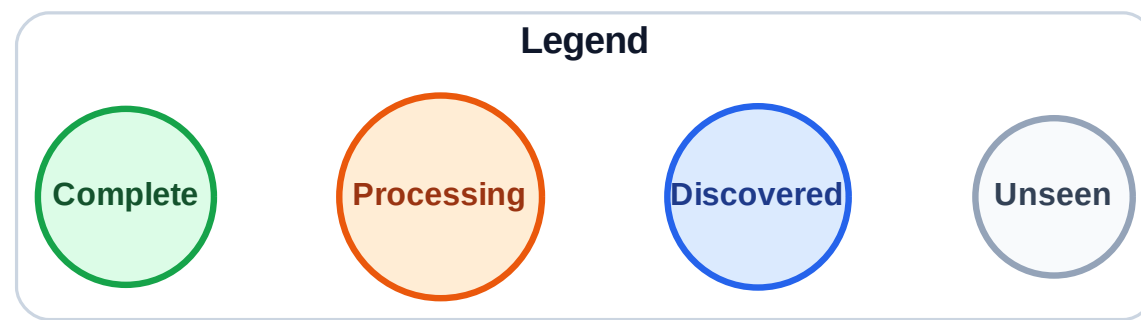
Stack (top at right): L | J | G | I | H | O

Visited: A, B, C, D, E, F, N



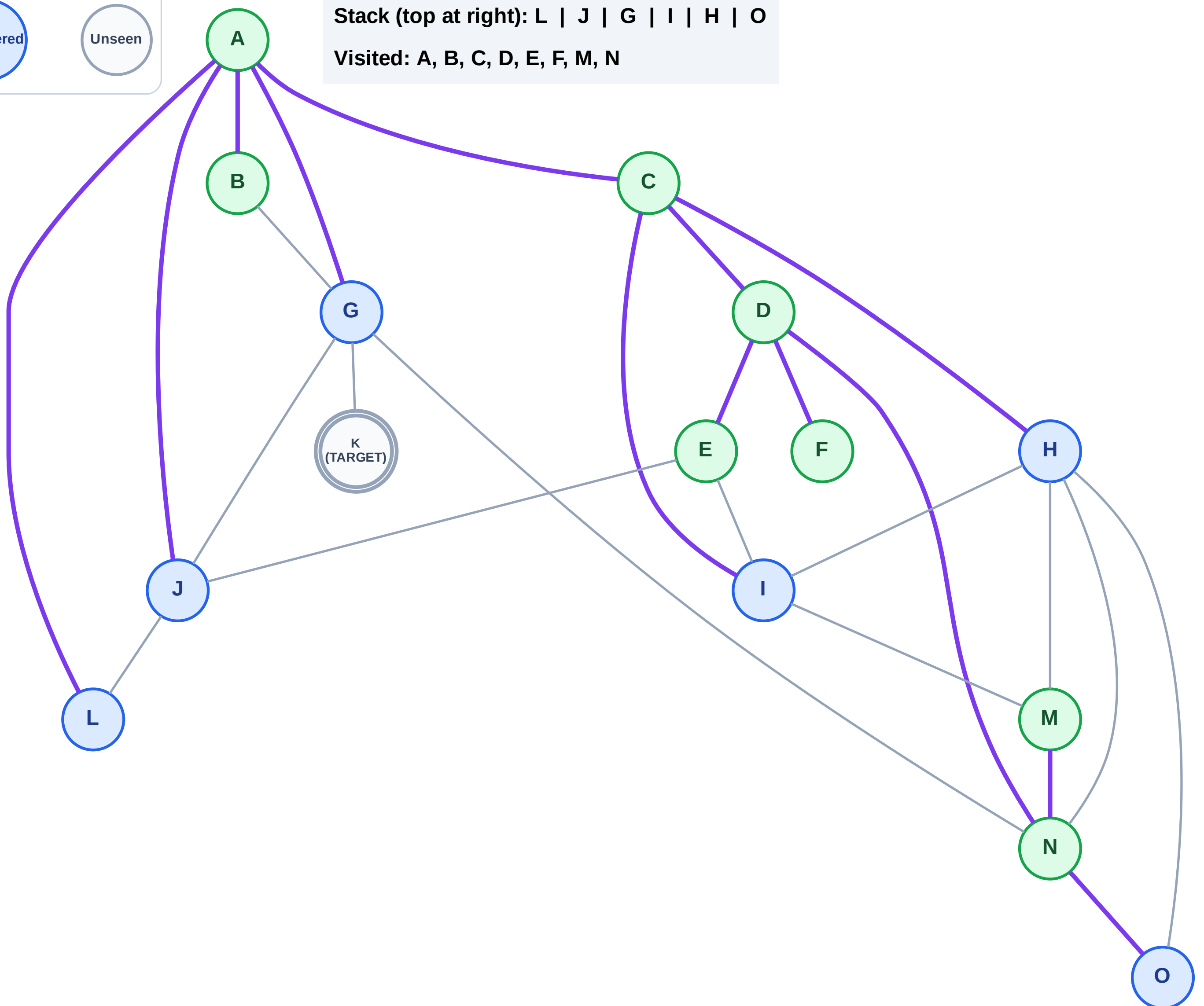
DFS Traversal

Step 17: No new neighbors from M



Stack (top at right): L | J | G | I | H | O

Visited: A, B, C, D, E, F, M, N



DFS Traversal

Step 18: Process node O

Legend

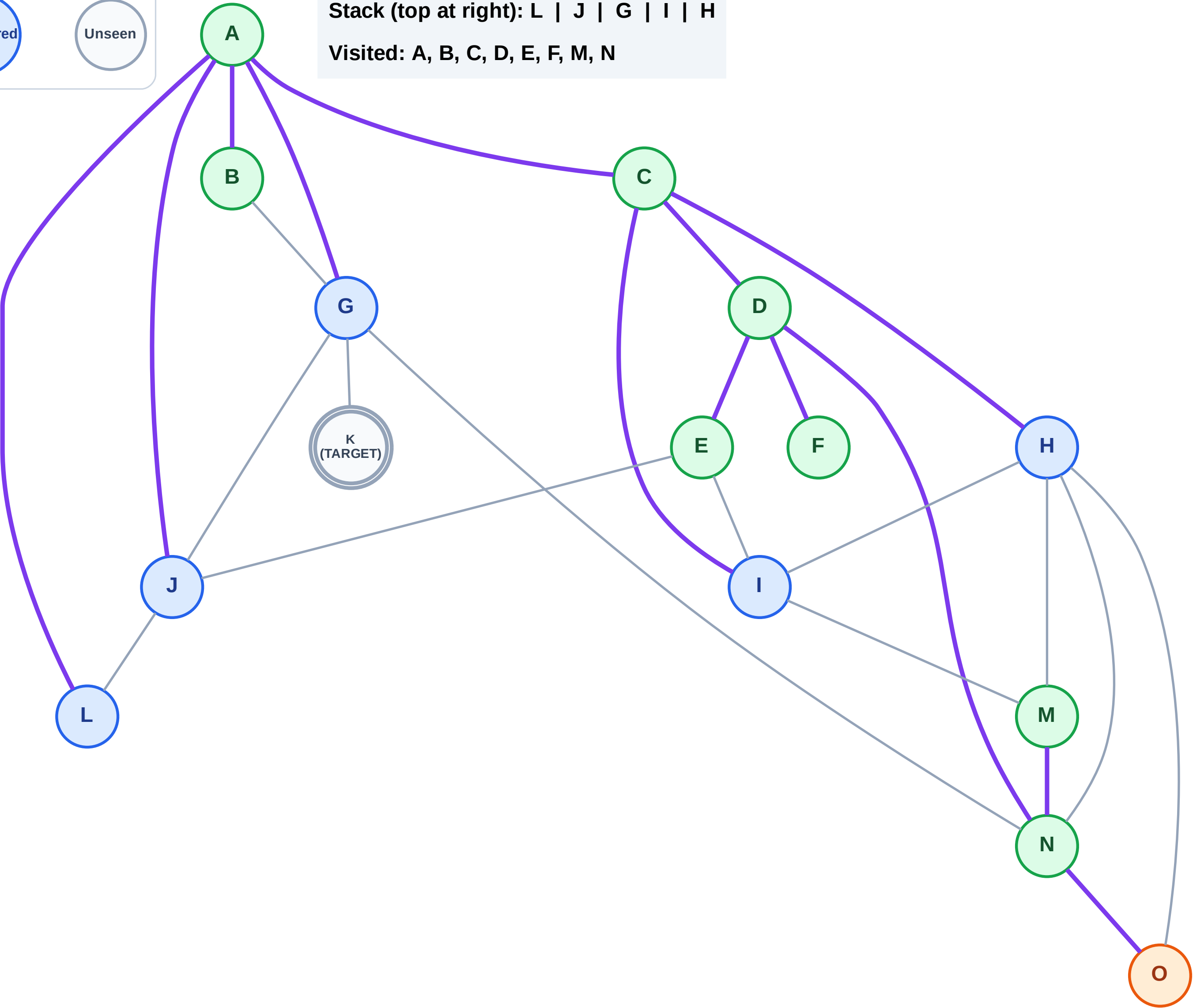
Complete

Processing

Discovered

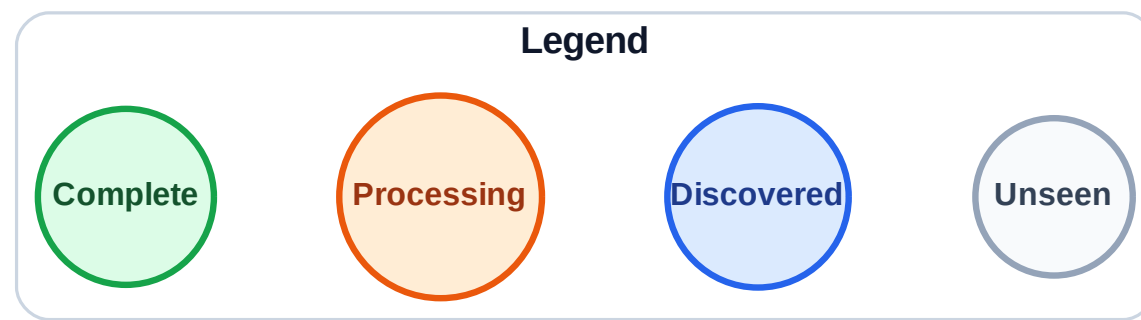
Unseen

Stack (top at right): L | J | G | I | H
Visited: A, B, C, D, E, F, M, N



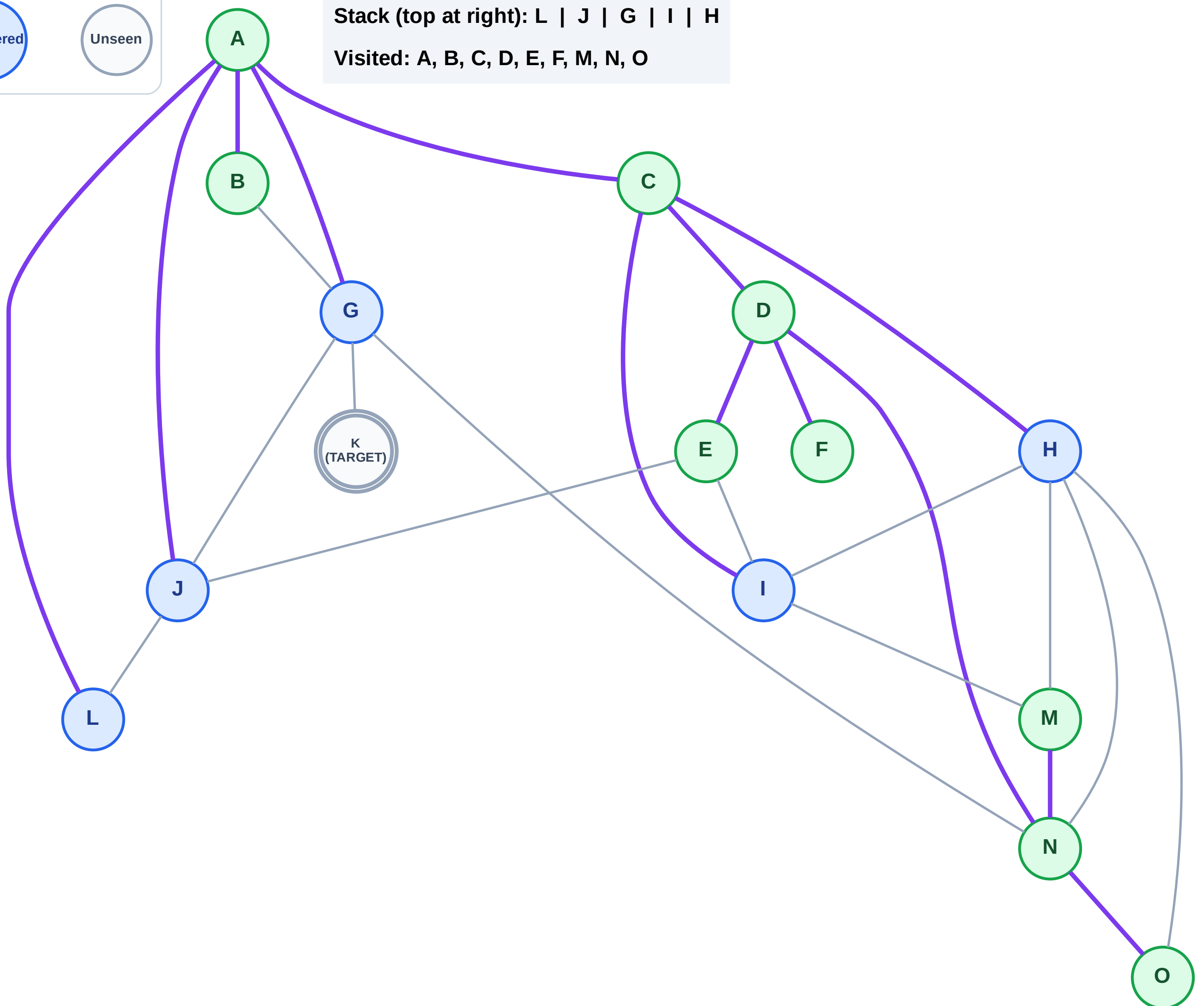
DFS Traversal

Step 19: No new neighbors from O

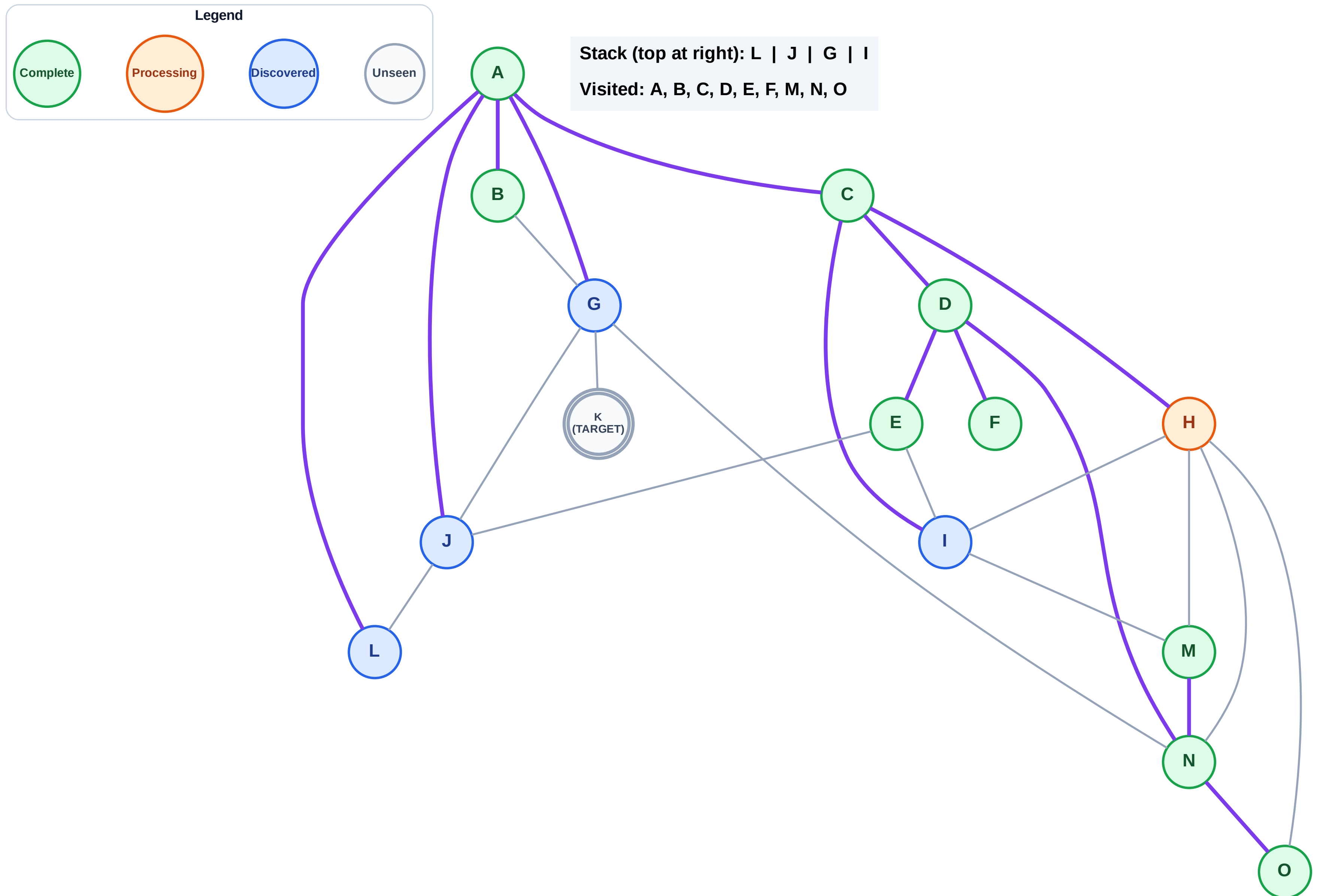


Stack (top at right): L | J | G | I | H

Visited: A, B, C, D, E, F, M, N, O



Step 20: Process node H



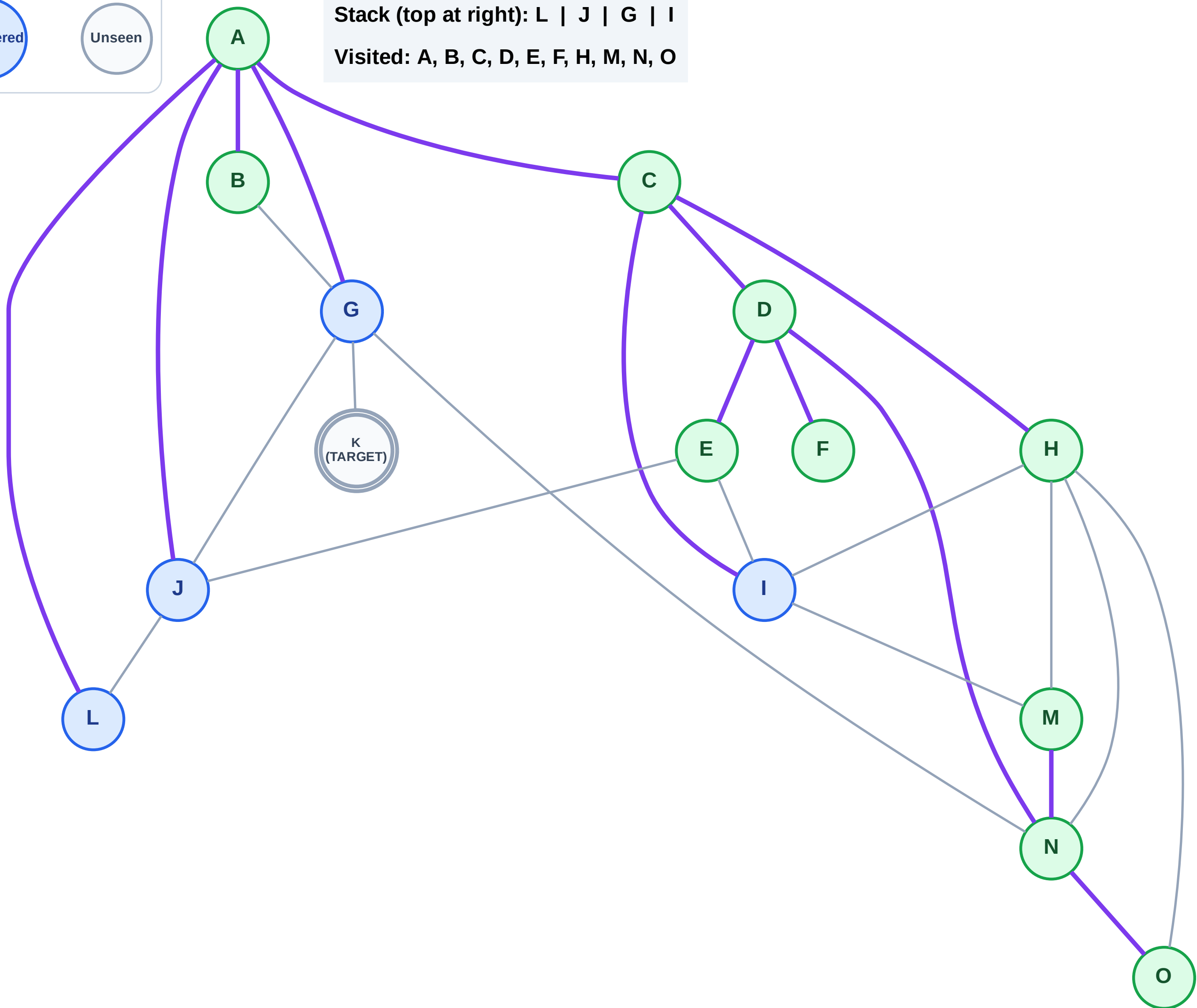
Step 21: No new neighbors from H

Stack (top at right): L | J | G | I

Legend

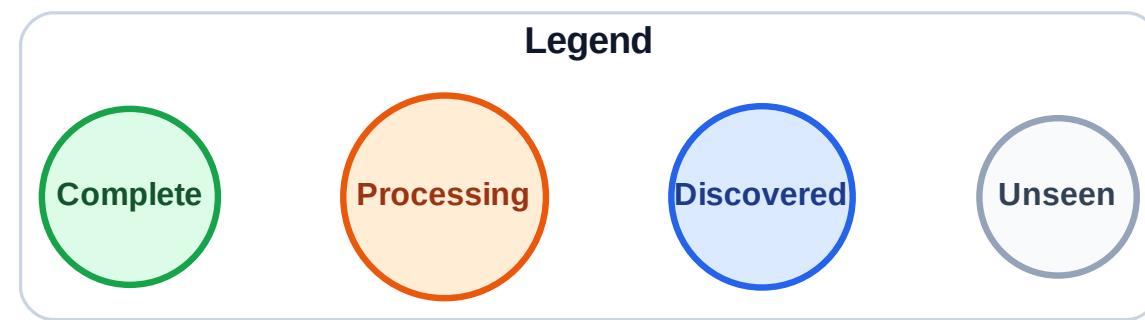
Processing

Unseen

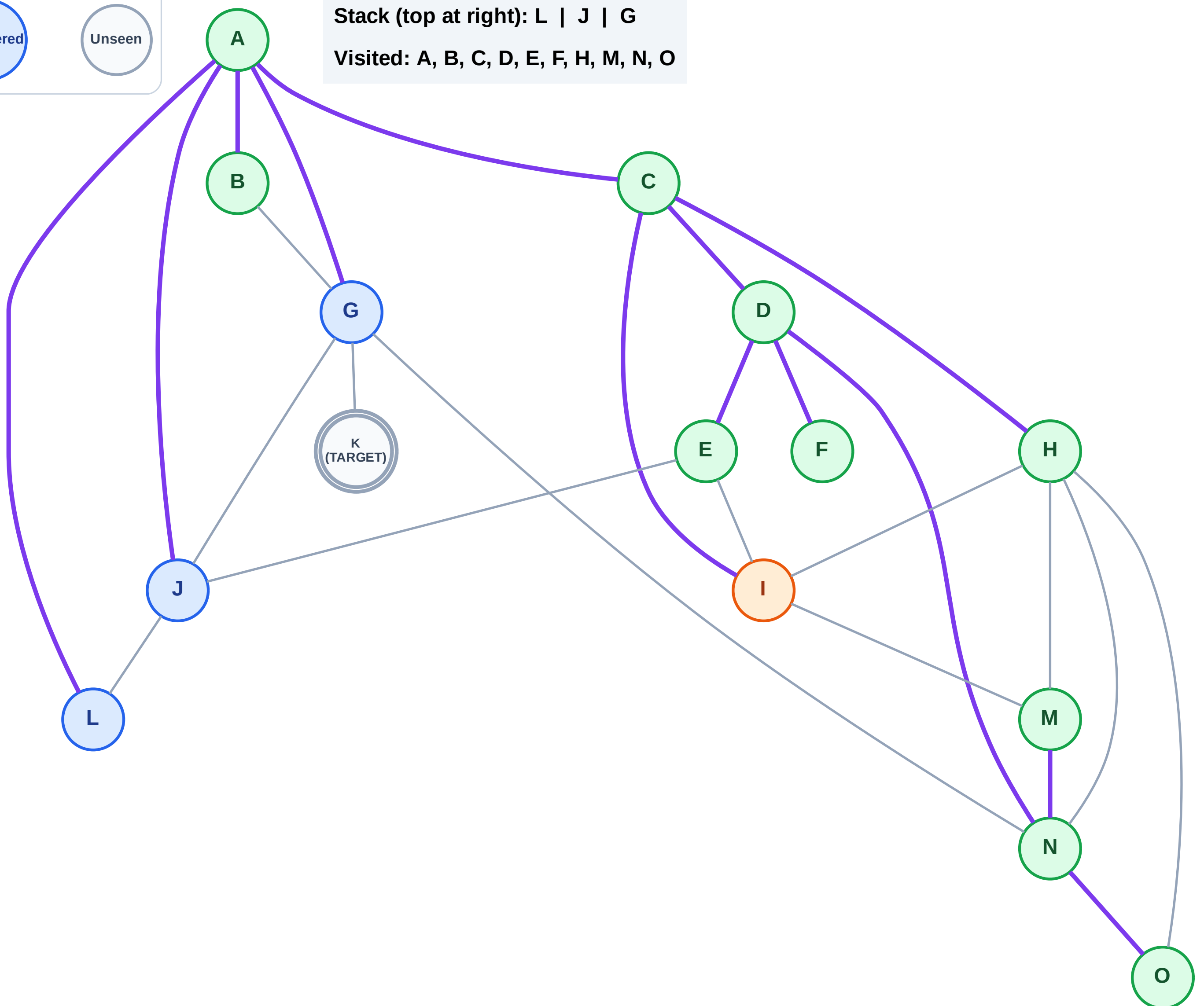


DFS Traversal

Step 22: Process node I

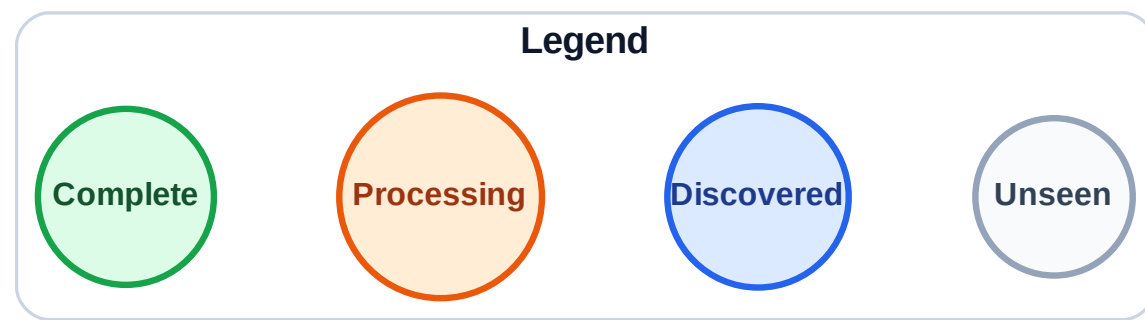


Stack (top at right): L | J | G
Visited: A, B, C, D, E, F, H, M, N, O



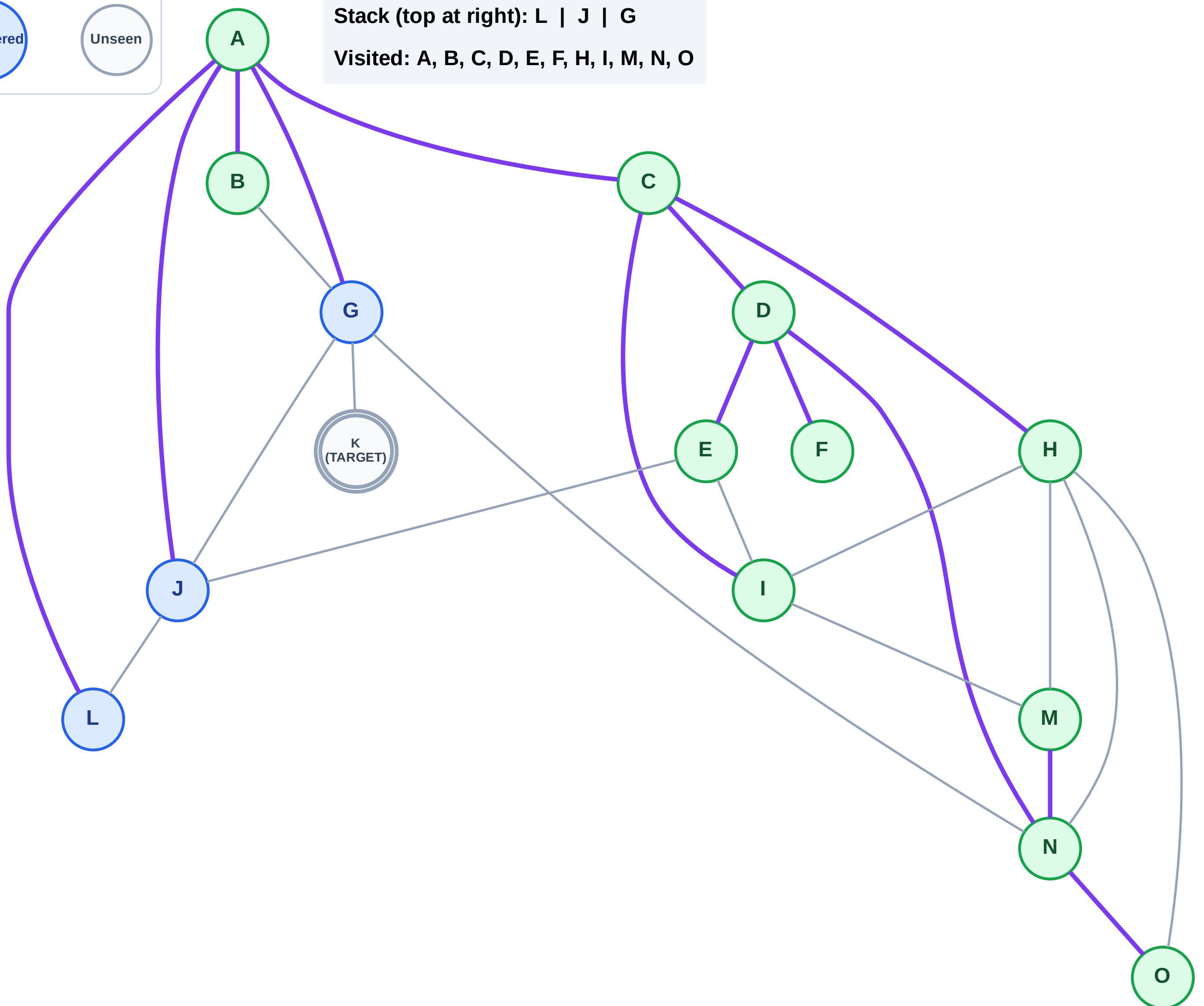
DFS Traversal

Step 23: No new neighbors from I



Stack (top at right): L | J | G

Visited: A, B, C, D, E, F, H, I, M, N, O



DFS Traversal

Step 24: Process node G

Legend

Complete

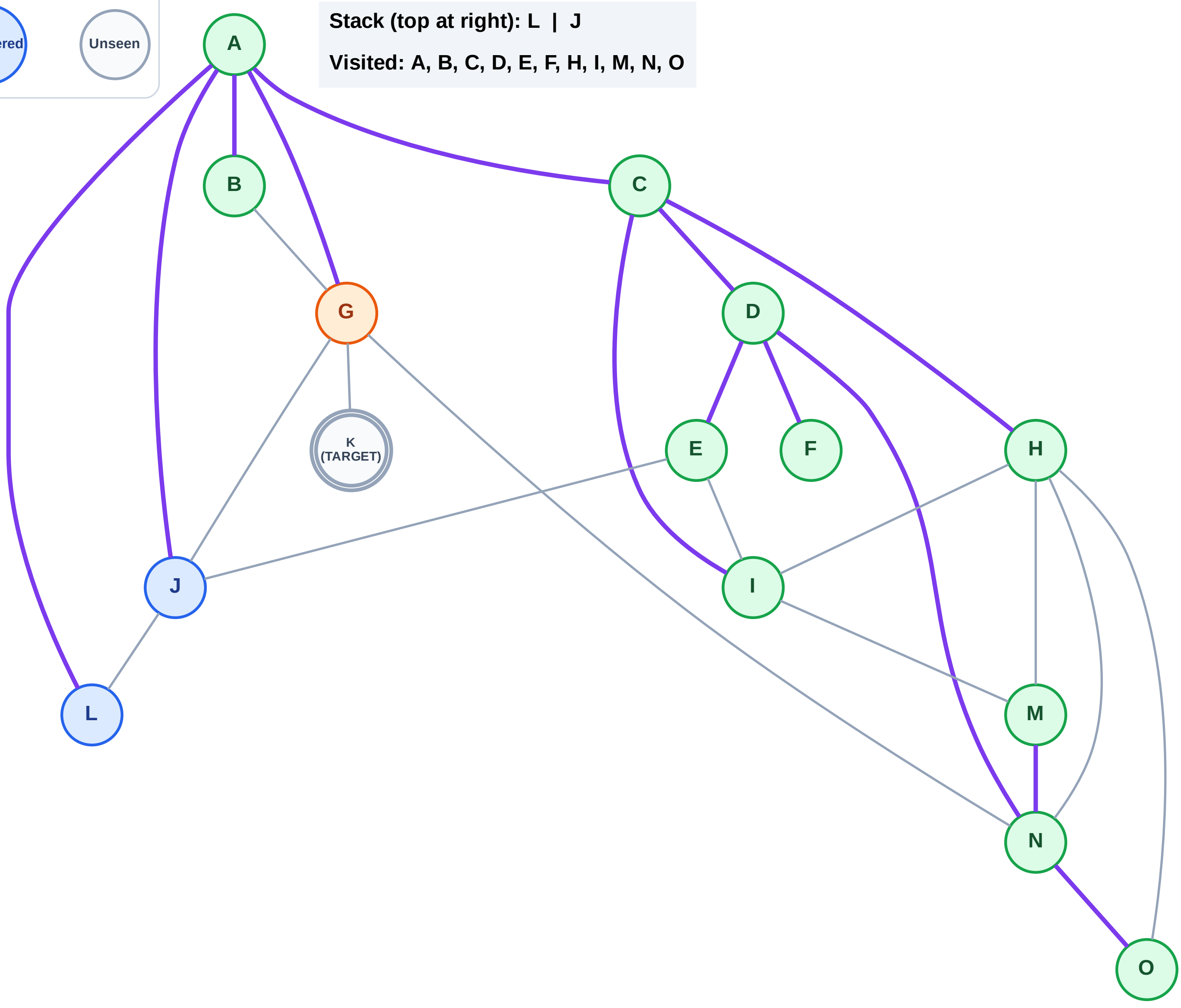
Processing

Discovered

Unseen

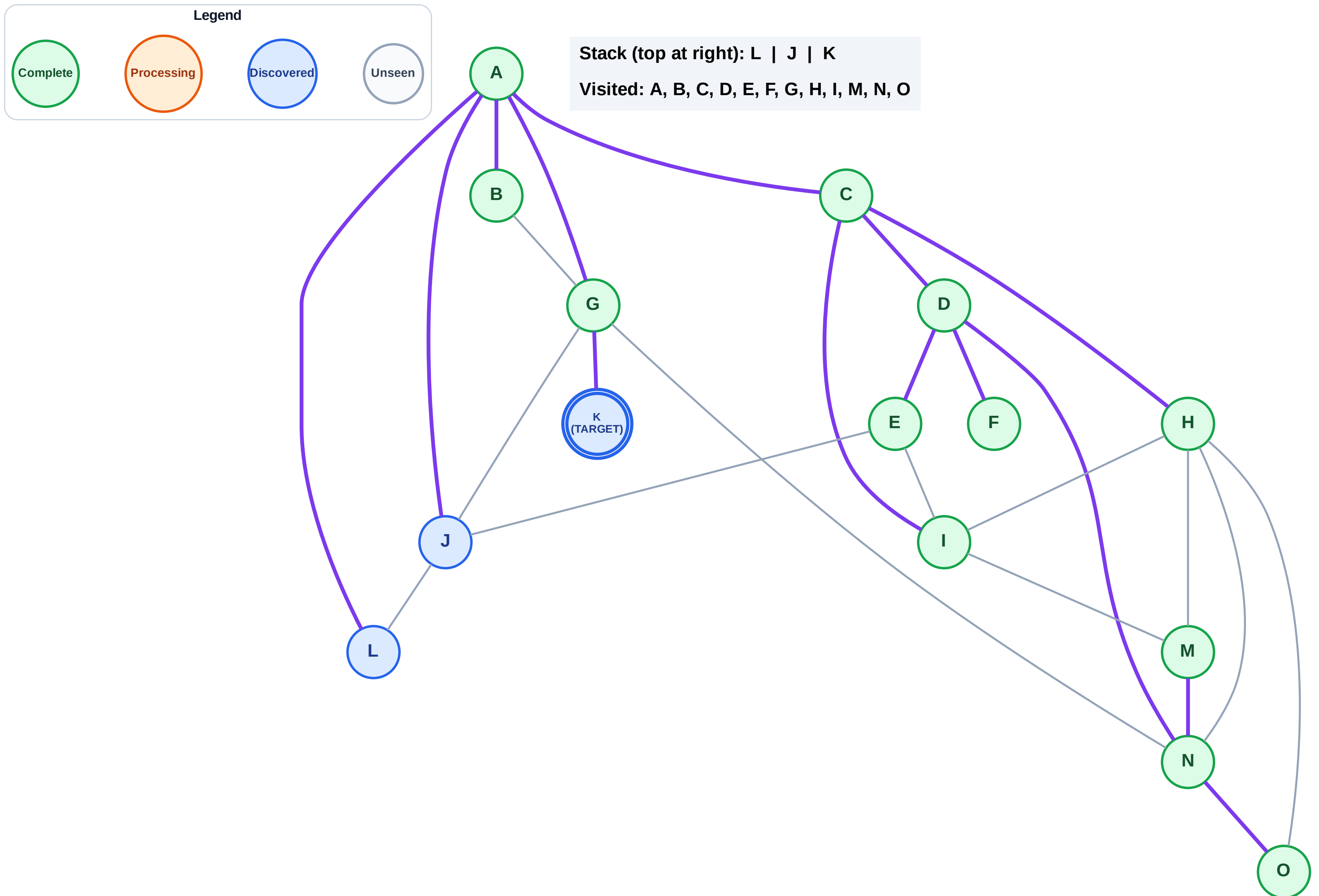
Stack (top at right): L | J

Visited: A, B, C, D, E, F, H, I, M, N, O



Step 25: Discover K from G

Step 25: Discover K from G



DFS Traversal

Step 26: Process node K

Legend

Complete

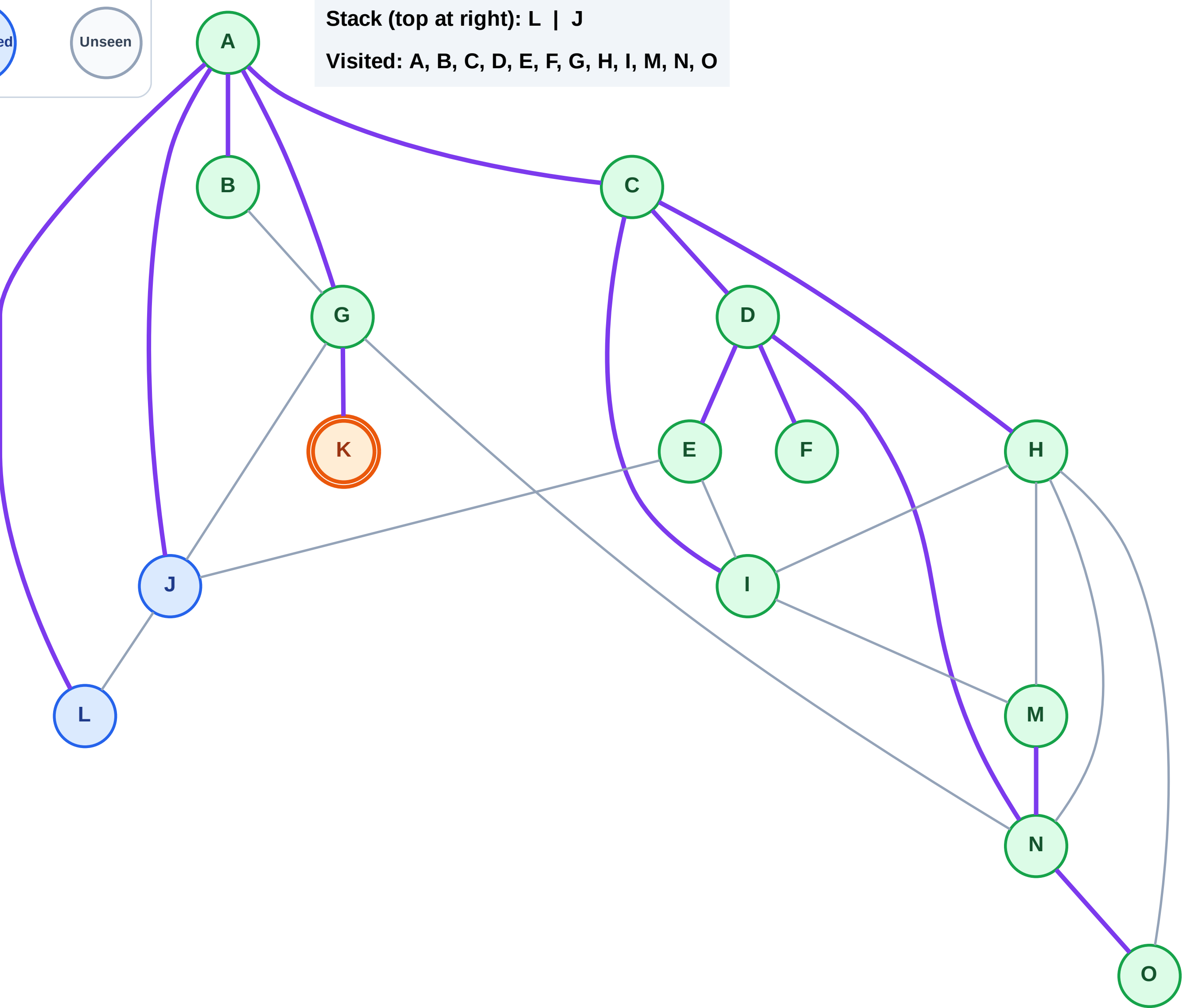
Processing

Discovered

Unseen

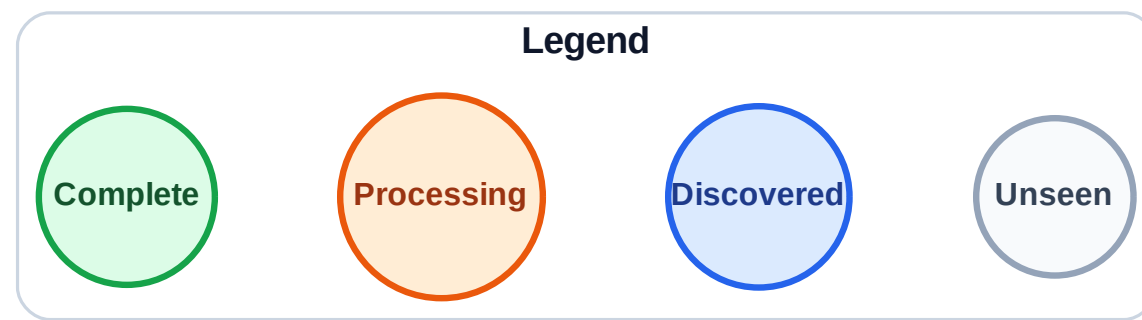
Stack (top at right): L | J

Visited: A, B, C, D, E, F, G, H, I, M, N, O



DFS Traversal

Step 27: TARGET NODE K REACHED! Path (2 hops): A → G → K



Stack (top at right): L | J

Visited: A, B, C, D, E, F, G, H, I, K, M, N, O

